

2

Straight-line Motion

In this chapter, we will learn about

- describing motion along a straight line
- representing motion with graphs
- analysing motion graphs
- equations of motion for studying uniformly accelerated motion
- free fall and the use of equations of motion to study such motion

2.1 Describing motion along a straight line

1 Expressing vectors in one dimension

Flipped classroom
Watch this and answer the questions.

In Chapter 1, we learned the meanings of displacement, velocity and acceleration. These are all vectors, meaning that each of them has a magnitude and a direction. The following map records a jogging session by Jenny (Fig 2.1a). Let us use these vectors to describe Jenny's motion.

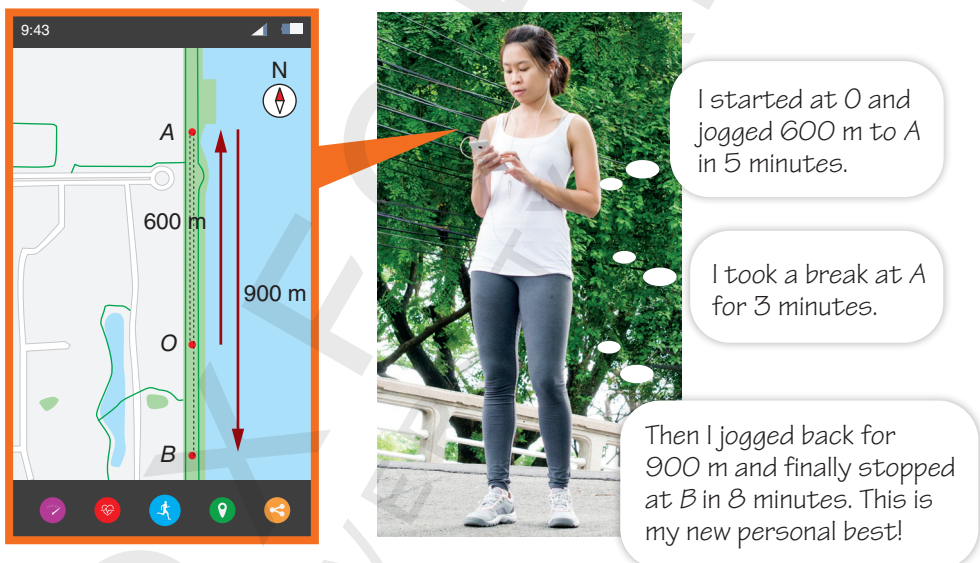


Fig 2.1a *Jenny's jogging path.*

Jenny jogged along a straight path. The displacement and velocity vectors can be in two opposite directions only. In this case, we can use the positive (+) and negative (–) signs to represent the directions of these vectors.

We may take either the north or the south as positive. Let us take the north as positive (Fig 2.1b). The opposite direction, the south, is therefore negative. Table 2.1a shows Jenny's displacements and average velocities.

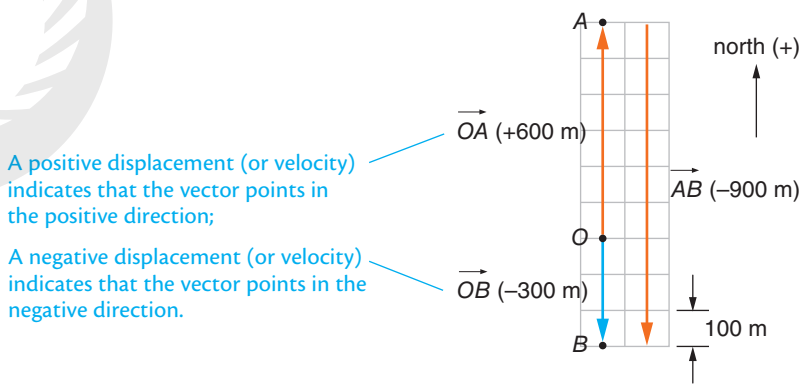


Fig 2.1b *Displacement vectors when the north is taken as positive.*

Journey	Displacement	Time taken	Average velocity
Jog from O to A	+600 m	5 min = 300 s	$\frac{+600}{300} = +2$ m s^{-1}
Rest at A	0	3 min = 180 s	0
Jog from A to B	−900 m	8 min = 480 s	$\frac{-900}{480} = -1.88$ m s^{-1}
Whole session	+600 + 0 + (−900) = −300 m	300 + 180 + 480 = 960 s	$\frac{-300}{960} = -0.313$ m s^{-1}

Table 2.1a *Displacements and average velocities of Jenny.*

Note the following:

1 Within any period of time, the sign of the average velocity is always the same as the sign of the displacement.

2 The sign of the instantaneous velocity tells the moving direction at that instant.

Example 1 Motion along a straight line

A cat walks on a straight fence. It first walks forwards for 10 s and the average speed is 0.6 m s^{-1}. Then it walks backwards for 20 s and the average speed is 0.45 m s^{-1}. Find its total displacement.



Solution

Tip
Always assign the positive direction at the beginning.

▶ Take forward as positive.

Tip
$v = \frac{s}{t} \Rightarrow s = vt$

▶ Displacement in first 10 s = $+0.6 \times 10 = 6$ m

Displacement in next 20 s = $-0.45 \times 20 = -9$ m

Total displacement = $+6 + (-9) = -3$ m

The total displacement is 3 m (backwards).

Learn by practice 1

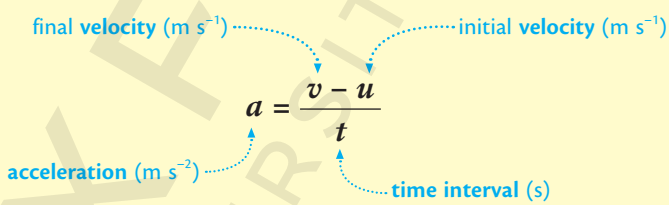
1 Tommy cycles along a straight road. He first cycles 1800 m towards north in 6 minutes. Then he cycles 960 m towards south in 4 minutes. Find his average speed and average velocity in these 10 minutes.

2 A woman walks forwards along a straight road at 0.8 m s^{-1} for 10 minutes. Then she jogs in the opposite direction at 3 m s^{-1} for 6 minutes. Find her total displacement and average velocity in these 16 minutes.

2 Acceleration along a straight line

a Average acceleration

Consider an object moving along a straight line. If its velocity changes from u to v in time t, its average acceleration is given by



Example 2 | Average acceleration of a lion

A lion is chased by a group of buffalos. It accelerates from rest to 12 m s^{-1} in 3 s along a straight line. Find its average acceleration.



Solution

Take the moving direction of the lion as positive.

$$a = \frac{v-u}{t} = \frac{12-0}{3} = 4 \text{ m s}^{-2}$$

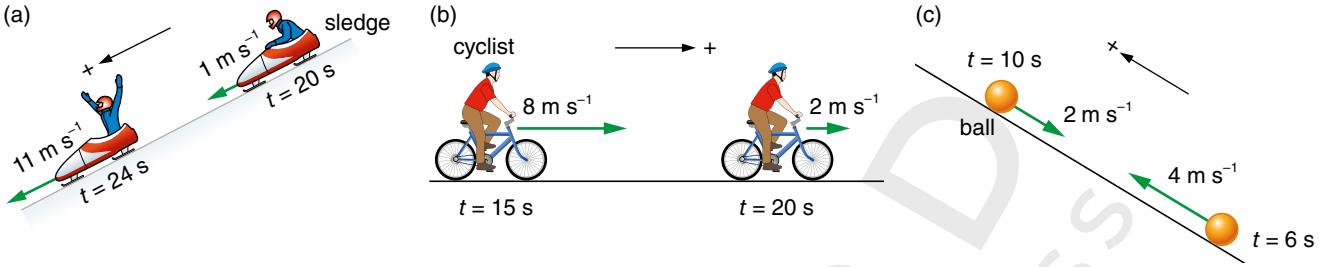
Its average acceleration is 4 m s^{-2} along its moving direction.

Quick check 1

→ E-version in OUP Exercise Platform is also available.

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1 Find the average acceleration of the object in motion in each of the following cases.



Simulation

→ This simulation shows the signs and magnitudes of velocity and acceleration in different motions. It is available on OUP web in PC version (.exe) only.

b Sign of acceleration

The sign of instantaneous velocity is the direction of motion at an instant. So what does the sign of acceleration tell us about the motion?

Consider two men running along a straight track (Fig 2.1c). One is running to the right while the other one is running to the left. Both men speed up from time $t = 0$ to $t = 2$ s and slow down from $t = 2$ s and $t = 6$ s.

The average acceleration in different time periods can be found. Let us see what happens when we **take rightwards as positive** (Table 2.1b).

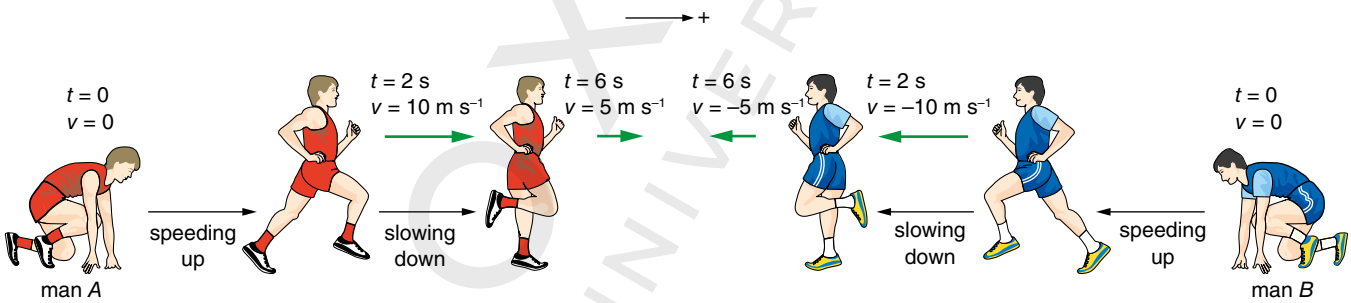


Fig 2.1c *Two men running along a straight track.*

Time	Average acceleration a	Motion	Sign of v	Sign of a
Man *A*				
0–2 s	$a = \frac{10 - 0}{2 - 0} = 5$ m s^{-2}	speeding up	+	+
2–6 s	$a = \frac{5 - 10}{6 - 2} = -1.25$ m s^{-2}	slowing down	+	–
Man *B*				
0–2 s	$a = \frac{-10 - 0}{2 - 0} = -5$ m s^{-2}	speeding up	–	–
2–6 s	$a = \frac{-5 - (-10)}{6 - 2} = 1.25$ m s^{-2}	slowing down	–	+

Table 2.1b *Motion of the men in different time periods.*

We can see that:

1 While each man keeps moving in the original direction, the sign of acceleration changes. This means that we **cannot** tell in what direction an object is moving only from the sign of acceleration.

2 We **cannot** tell whether an object is speeding up or slowing down only from the sign of acceleration.

3 When the acceleration and the velocity of a moving object have the same sign (i.e. the same direction), the object is speeding up. Otherwise, it is slowing down (Table 2.1c). This is always true no matter whether the object is moving in the positive or negative direction.

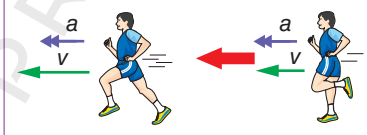
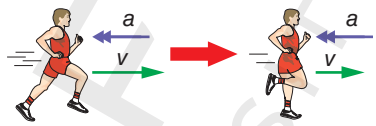
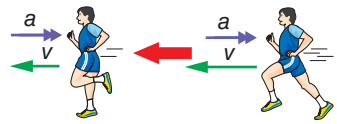
speeding up (a and v have the same sign)		
slowing down (a and v have opposite signs)		

Table 2.1c *The velocity and acceleration vectors when speeding up or slowing down.*

- **The sign of acceleration of an object does not tell the direction of motion, nor whether the object is speeding up or slowing down.**
- **The object is speeding up if its velocity v and acceleration a have the same sign; it is slowing down if v and a have opposite signs.**

As $a = -2$ m s^{-2}, v decreases by 2 m s^{-1} in every second. ▶

Now consider a car initially moving at 8 m s^{-1} forwards slows down uniformly at 2 m s^{-2} on a straight road. Figure 2.1d shows how it will move in the next 3 s.

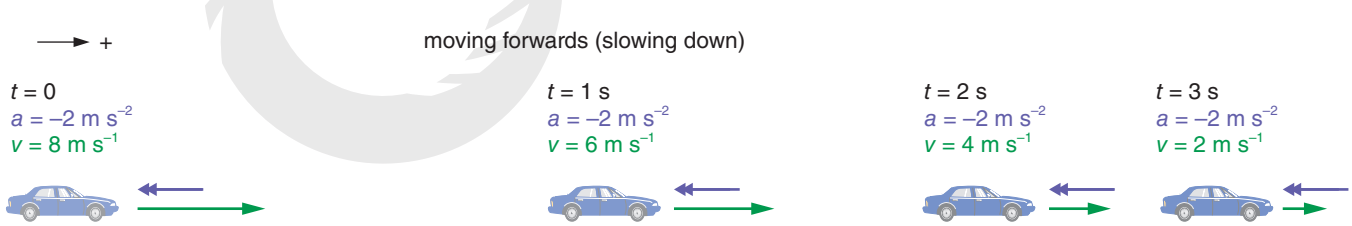


Fig 2.1d *A car slowing down uniformly.*

Calculation involving uniform acceleration (e.g. finding v at t = 6.5 s) will be discussed in Ch 2.3.

▶ What happens if the car keeps moving with this acceleration? As v further decreases, it will become negative, meaning that the car will change its moving direction (Fig 2.1e).

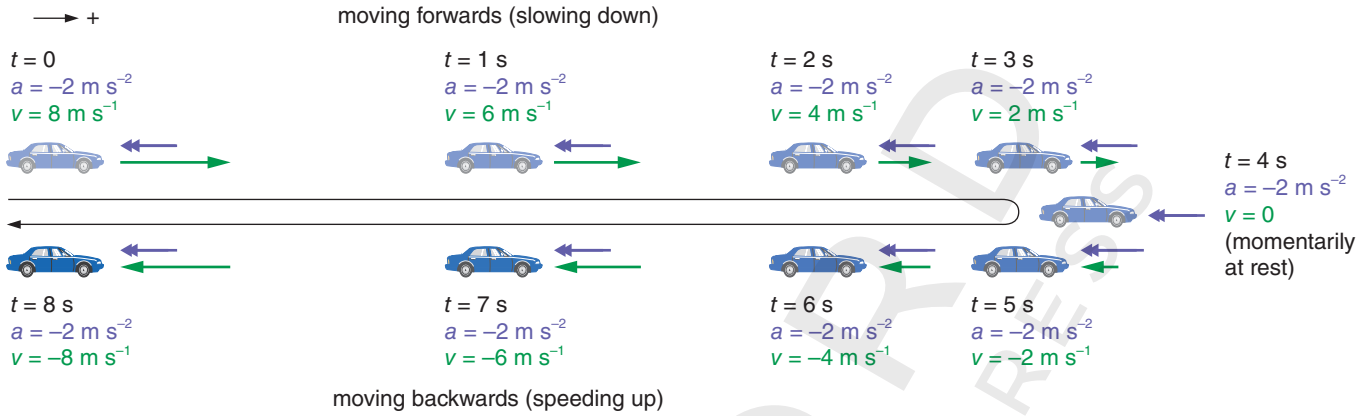


Fig 2.1e *A car with an acceleration opposite to its initial velocity eventually changes its moving direction.*

From Figure 2.1e, we can see that:

1 At t = 4 s, **the speed becomes zero**. This means the car is **momentarily at rest**.

2 Then the velocity becomes negative, meaning that it moves in a backward direction. In other words, the car changes its moving direction.

3 After t = 4 s, the velocity becomes more and more negative. In this case, since **the direction of acceleration is the same as the direction of velocity**, the car **speeds up**. Its speed increases by 2 m s^{-1} each second.

Quick check 2

→ E-version in OUP Exercise Platform is also available.

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1 A car is speeding up along a straight line. Which of the following must be correct?

A The car is moving in a positive direction.

B The car has a positive acceleration.

C The direction of the car's acceleration is the same as the direction of its velocity.

2 Which of the following is the condition for an object to be momentarily at rest?

A The speed of the object is zero at a moment.

B The acceleration of the object is zero at a moment.

C Both the speed and acceleration of the object are zero at a moment.

3 A ball is projected up a slope from point X. After reaching the highest point Y, it moves back to X (Fig a).

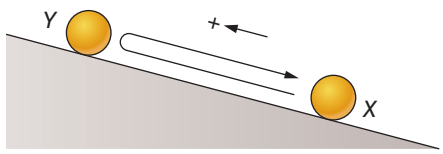


Fig a

Fill in the blanks with '+', '0' or '–' to show the direction of acceleration in different parts of the journey. Take the direction up the slope as positive.

From X to Y: _____

At Y: _____

From Y to X: _____

Exercise 2.1

1 An object moves along a straight line and has a positive acceleration. Which of the following statements about this object must be correct?

A It is speeding up.

B It is slowing down.

C Its velocity is changing.

D Its moving direction will change.

***2** A working platform (Fig a) is 20 m above the ground. It moves downwards at a constant speed for 2 minutes to the ground. Then it moves upwards at the same speed for 2.5 minutes. Take downward as positive.



Fig a

What is the total displacement of the working platform over the whole journey?

A −25 m

B −5 m

C 5 m

D 25 m

***3** A runner is running along a straight line. Take leftward as positive. If the velocity of the runner is negative while his acceleration is positive, which of the following best describes his motion?

A Running to the left with decreasing speed

B Running to the right with decreasing speed

C Running to the left with increasing speed

D Running to the right with increasing speed

4 A bus slows down from 36 km h^{-1} to a stop in 5 s on a straight road. What is its average deceleration?

5 The figure below shows the positions of a car on a straight road at different instants (Fig b). The car is in uniform motion during this period of time.



Fig b

Take forward as positive.

(a) What is the velocity of the car at t = 1 s? You may consider the positions of the car's front end in doing calculation.

(b) Then the car accelerates starting from t = 2 s. Its velocity becomes 8 m s^{-1} towards the left at t = 4 s. Find its average acceleration from t = 2 s to t = 4 s.

***6** A ball is thrown upwards with an initial speed of 8 m s^{-1} (Fig c). It accelerates uniformly at 10 m s^{-2} downwards during its flight. What is its velocity 2 s after it is thrown?



Fig c

2.2 Motion graphs

Data of a motion presented as graphs helps us better understand the entire motion. In this part, we will learn three types of motion graphs.

1 Motion graphs of straight-line motion

a Displacement–time graph (*s*–*t* graph)

DSE goal

Present information on *s*–*t* graphs.

Use *s*–*t* graphs to determine the displacement and velocity.

DSE exam

18(1A)Q7

Simulation

→ This simulation shows motion represented by *s*–*t* graphs. Ss can choose a graph and then observe the motion. Conversely, they can observe a motion and then sketch an *s*–*t* graph to represent it.

A **displacement–time graph** (*s*–*t* graph) shows the **displacement of an object from a reference point** at different instants. The *x*-axis is the time axis while the *y*-axis is the displacement axis.

Suppose a helium balloon moves upwards with a constant velocity of 2 m s^{-1}, starting from a position 1 m above the ground at time $t = 0$ (Fig 2.2a). We can take upwards as positive and record its displacement s from the ground (the reference point) at different instants in a table. If we plot the data in a graph, we get the balloon's *s*–*t* graph.

Time *t* / s	Displacement *s* from ground / m
0	+1
1	+3
2	+5
3	+7

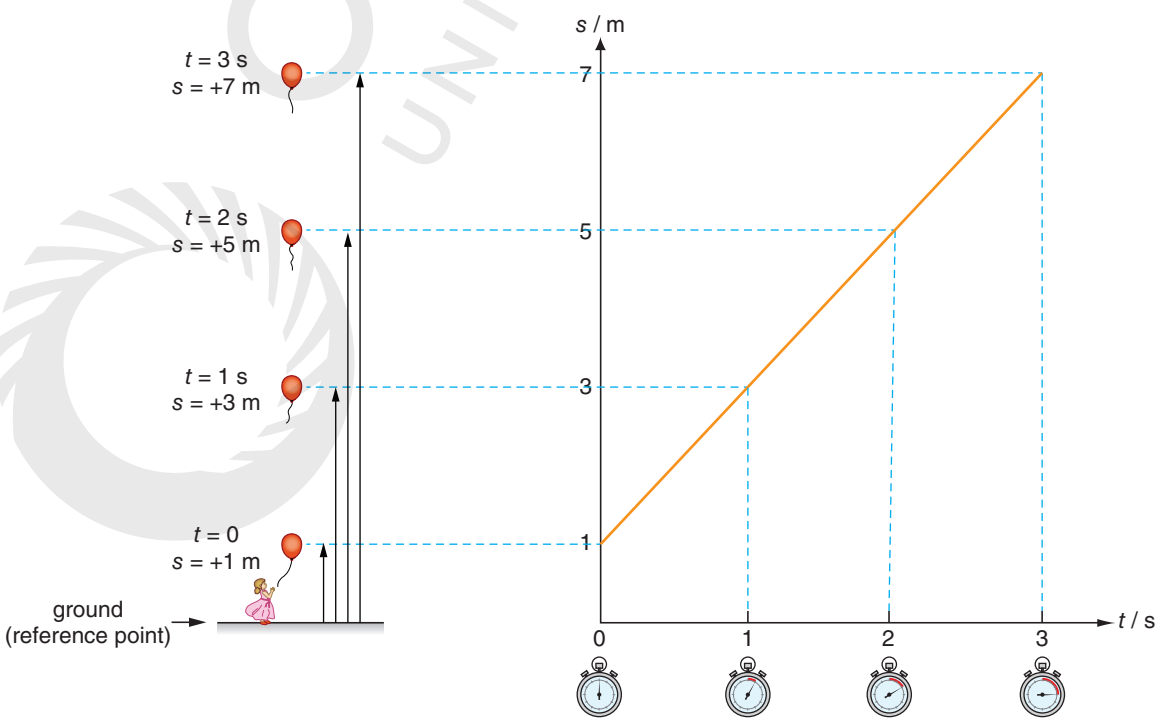


Fig 2.2a *How the s–t graph of a helium balloon is plotted.*

displacement–time graph 位移—時間關係線圖

We defined velocity as displacement per unit time on p.14. The displacement there refers to the change in position within the time period. No reference point is assigned.

We can see that s here actually tells the **position of the balloon from the ground at an instant**. It is **not the displacement** (i.e. the change in position) **of the balloon from $t = 0$ to that instant** (Fig 2.2a on p.43).

Furthermore, the slope of the graph gives the velocity of the object (Fig 2.2b).

$$\text{Slope of } s\text{–}t \text{ graph} = \frac{\Delta s}{\Delta t} = \text{velocity}$$

Delta (Δ) is used to denote a change of a quantity. Δs and Δt mean the change in s and the change in t respectively.

Unit of slope $= \frac{\text{m}}{\text{s}}$
$= \text{m s}^{-1}$
= unit of velocity

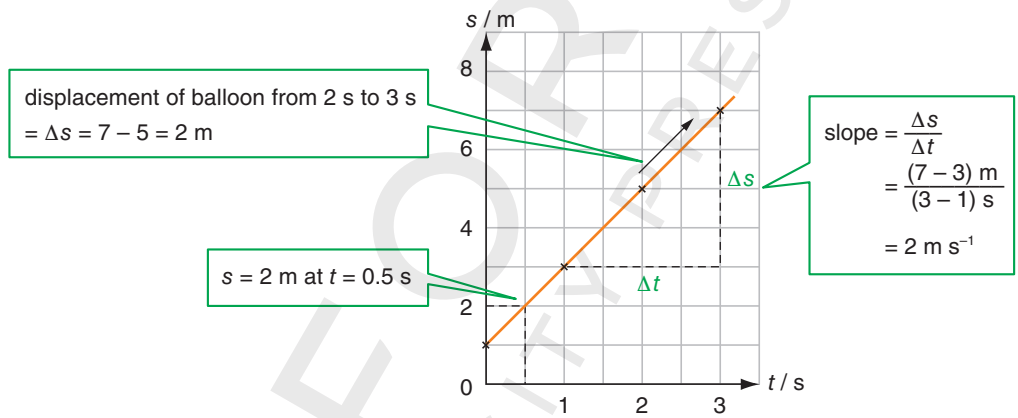


Fig 2.2b *How position (s), displacement (Δs) and average velocity (Δs/Δt) are found from an s–t graph.*

From this relation, the following can be obtained from an s–t graph.

1. If the graph is a **straight line** (constant slope), the object moves with a **constant velocity**, i.e. it is in **uniform motion**.
2. A **steeper slope** represents a velocity with a larger magnitude, i.e. a **higher speed** (Fig 2.2c). **Zero slope** means zero velocity, i.e. **at rest**.
3. Displacement may decrease (or become more negative). In this case, the s–t graph will have a negative slope. The **sign of the slope** gives the **direction of velocity** (Fig 2.2d). Moreover, a more negative slope means the object moving with a higher speed in the negative direction (Fig 2.2e).

Skill *Mathematics*
Slope ▸ p.452

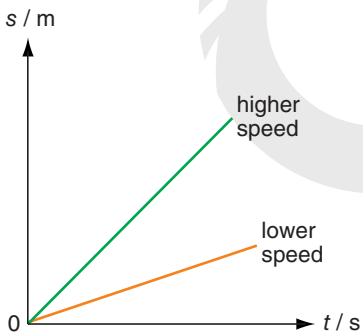


Fig 2.2c *A steeper slope represents a higher speed.*

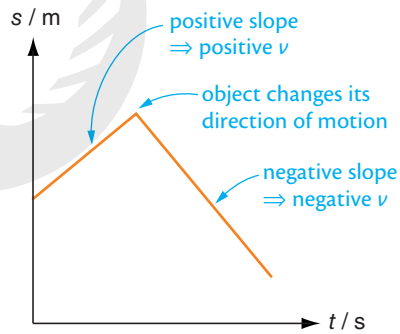


Fig 2.2d *Different signs of slope represent different directions of velocity.*

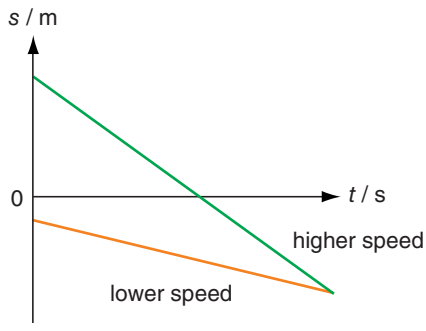
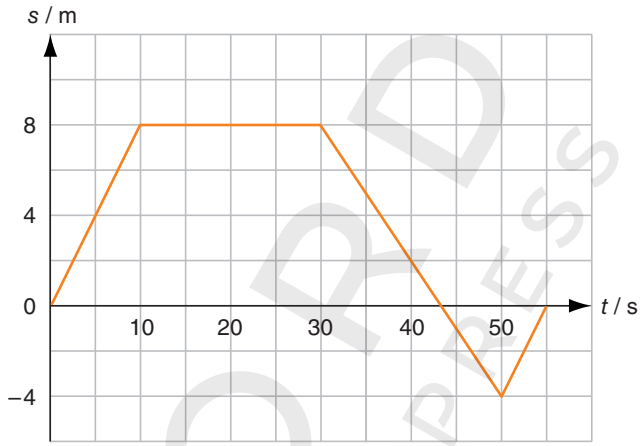


Fig 2.2e *A more negative slope represents a higher speed in the negative direction.*

Example 3 | Analyzing an *s*–*t* graph

The *s*–*t* graph of a girl walking along a straight road is as shown.



(a) During which time periods is the girl walking in the positive direction?

(b) Find

(i) the displacement of the girl and

(ii) the distance travelled by the girl

from t = 5 s to t = 40 s.

(c) Find the velocities of the girl at t = 5 s and t = 40 s.

(d) Find the average velocity of the girl from t = 0 to t = 50 s.

Solution

(a) During 0–10 s and 50–55 s

(b) (i) Displacement

$= 2 - 4 = -2$ m

(ii) Distance travelled

$= (8 - 4) + (8 - 2) = 10$ m

(c) Apply velocity = slope of *s*–*t* graph.

At t = 5 s,

$$\text{velocity} = \frac{8 - 0}{10 - 0} = 0.8 \text{ m s}^{-1}$$

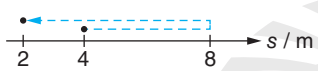
At t = 40 s,

$$\text{velocity} = \frac{(-4) - 8}{50 - 30} = -0.6 \text{ m s}^{-1}$$

(d) Average velocity $= \dfrac{(-4) - 0}{50 - 0} = -0.08 \text{ m s}^{-1}$

Tip

A sketch of the girl's motion from t = 5 s to t = 40 s:



Tip

Constant velocity from t = 0 to t = 10 s.

Tip

Constant velocity from t = 30 s to t = 50 s.

Tip

$\dfrac{s}{t} = \dfrac{\text{total displacement}}{\text{total time of travel}}$

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Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 A boy walks along a straight road. His s–t graph is shown in Figure a.

(a) Complete Table a.

(b) Find the total displacement and average velocity of the boy from $t = 0$ to $t = 500$ s.

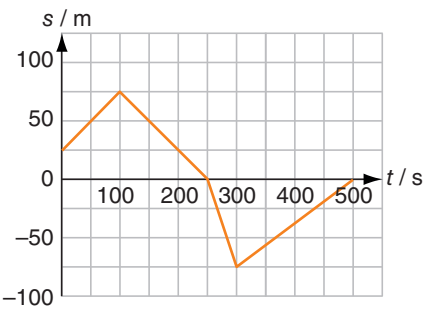


Fig a

Time interval	Displacement / m	Velocity / m s^{-1}
0–100 s	+50	
100–250 s		
250–300 s		
300–500 s		

Table a

2 Figure b shows the s–t graph of an object.

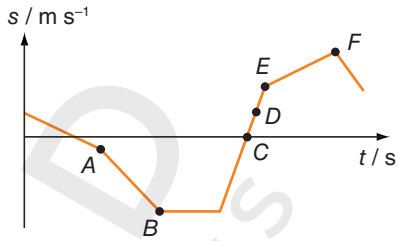


Fig b

Determine each of the following motions of the object is represented by which point (A–F) in the figure.

(a) Changing to a lower speed

(b) Changing to a higher speed

(c) Changing its direction of motion

(d) Passing the reference point

(e) Passing its position at $t = 0$ again

(f) Starting to rest

Learn by practice 2

1 Figure a shows the s–t graph of a car moving on a straight road. s is taken as the displacement from a tree.

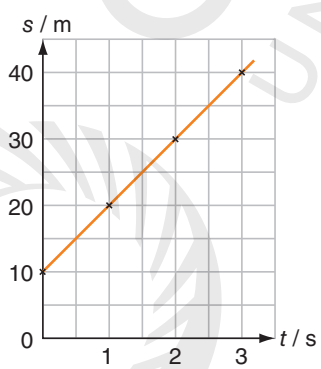


Fig a

(a) How far is the car from the tree at $t = 0.5$ s?

(b) What is the displacement of the car during 2–3 s?

(c) In Figure a, sketch an s–t graph of a car moving with a larger constant speed from the same position at $t = 0$.

2 The s–t graphs of cars A and B travelling along a straight road are shown in Figures b and c respectively.

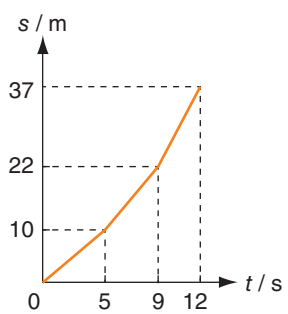


Fig b

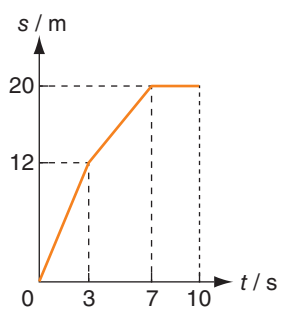


Fig c

(a) Find the velocities of each car at $t = 2$ s, 6 s and 10 s.

(b) Which car has sped up during the journey? Which car has slowed down?

Writing practice

Question

Refer to the figure in Example 3 (p.45). Describe the motion of the girl from $t = 0$ to $t = 50$ s. (3 marks)

Answer

During 0–10 s, the girl moves in the positive direction with a constant speed. 1A

During 10–30 s, the girl is at rest. 1A

During 30–50 s, the girl moves in the negative direction with a constant speed. 1A

Tip

You may describe the motion of an object by following the steps below.

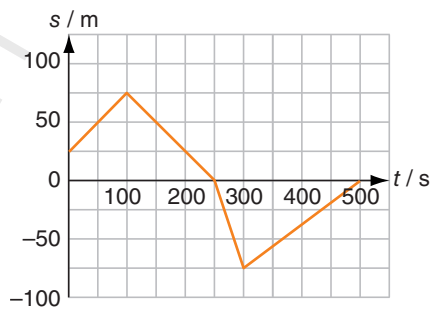
① Divide the motion into different stages.

② In each stage, what is the direction of motion? Or is the object at rest?

③ If the object is in motion, does it speed up / slow down / move with constant speed?

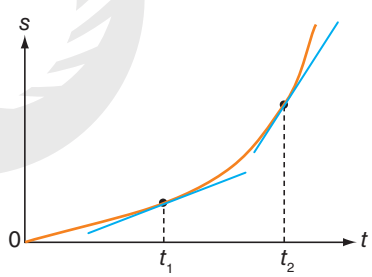
Practise yourself

1 A boy walks along a straight road. His s–t graph is shown on the right. Describe the motion of the boy from $t = 0$ to $t = 500$ s. (4 marks)



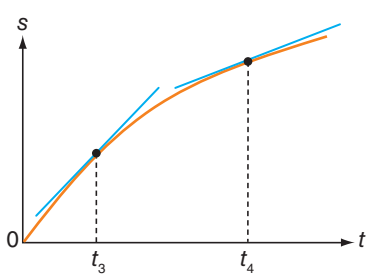
When an s–t graph is a curve, the velocity of the object is changing. The velocity at a particular instant, i.e. the instantaneous velocity, is given by the slope of the curve at that instant. Figures 2.2f and 2.2g show two examples. The slope of the graph in Figure 2.2f becomes steeper and steeper, showing that the object speeds up. On the contrary, the graph in Figure 2.2g shows that the object slows down.

Ss need not calculate the instantaneous velocity from an s–t graph which is a curve. However, they should know how the velocity changes from the slope of the curve.



slope at t_2 > slope at t_1
⇒ velocity at t_2 > velocity at t_1

Fig 2.2f *s–t graph of an object which speeds up.*



slope at t_4 < slope at t_3
⇒ velocity at t_4 < velocity at t_3

Fig 2.2g *s–t graph of an object which slows down.*

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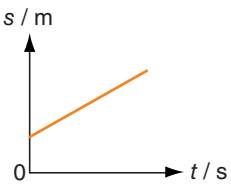
Google Forms Microsoft Forms

Quick check 4

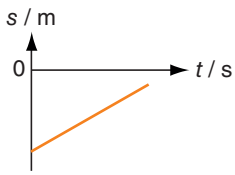
→ E-version in OUP Exercise Platform is also available.

1 Four *s*–*t* graphs are shown below.

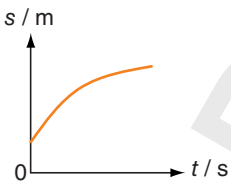
Graph I



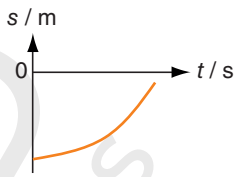
Graph II



Graph III



Graph IV



Determine which graph(s) can represent each of the following motions. For each motion, you may choose more than one graph.

(a) Moving in the positive direction

(b) Moving in the negative direction

(c) Speeding up

(d) Moving at a constant velocity

(e) Slowing down

→ This simulation shows motion represented by *v*–*t* graphs. Ss can choose a graph and then observe the motion. Conversely, they can observe a motion and then sketch a *v*–*t* graph to represent it.

Simulation

b Velocity–time graph (*v*–*t* graph)

DSE goal

Present information on *v*–*t* graphs.

Use *v*–*t* graphs to determine the displacement, velocity and acceleration.

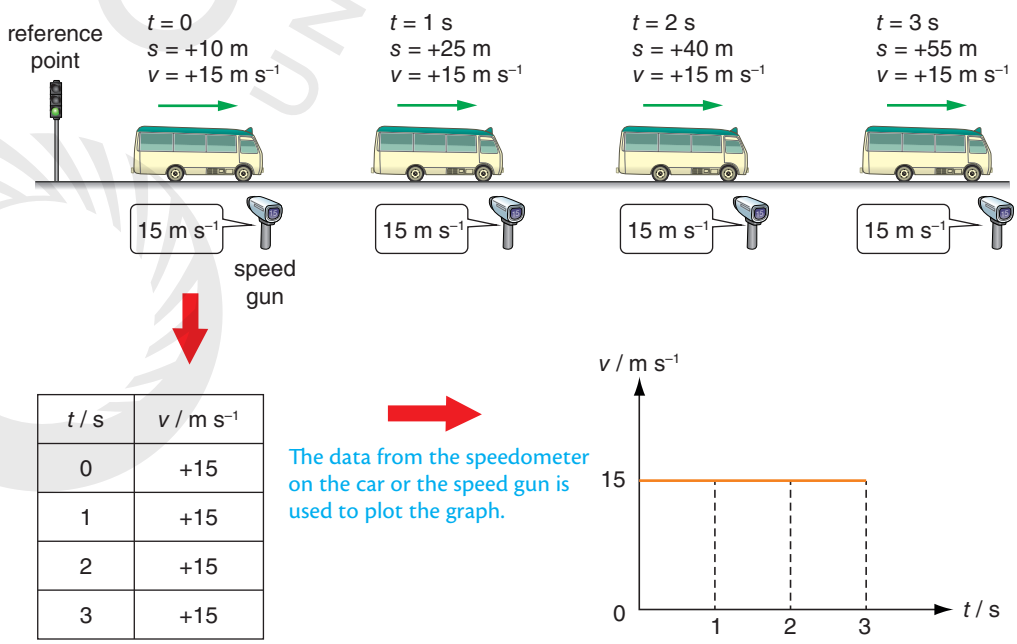
Present information on *a*–*t* graphs.

DSE exam

14(1B)Q3, 15(1B)Q4(a)–(b), 15(1A)Q9, 18(1A)Q10, 18(1A)Q11, 21(1A)Q7, 23(1B)Q3(c)(i), etc.

A **velocity–time graph** (*v*–*t* graph) is a graph that shows the velocity of an object at different instants.

Consider the motion of a car moving along a straight road with a **constant velocity** of +15 m s^{-1}, starting from a position at 10 m from the reference point at time t = 0 (Fig 2.2h). The car's ***v*–*t* graph is a horizontal line**.



t / s	*v* / m s^{-1}
0	+15
1	+15
2	+15
3	+15

The data from the speedometer on the car or the speed gun is used to plot the graph.

Fig 2.2h *Plotting the v–t graph of a car in uniform motion.*

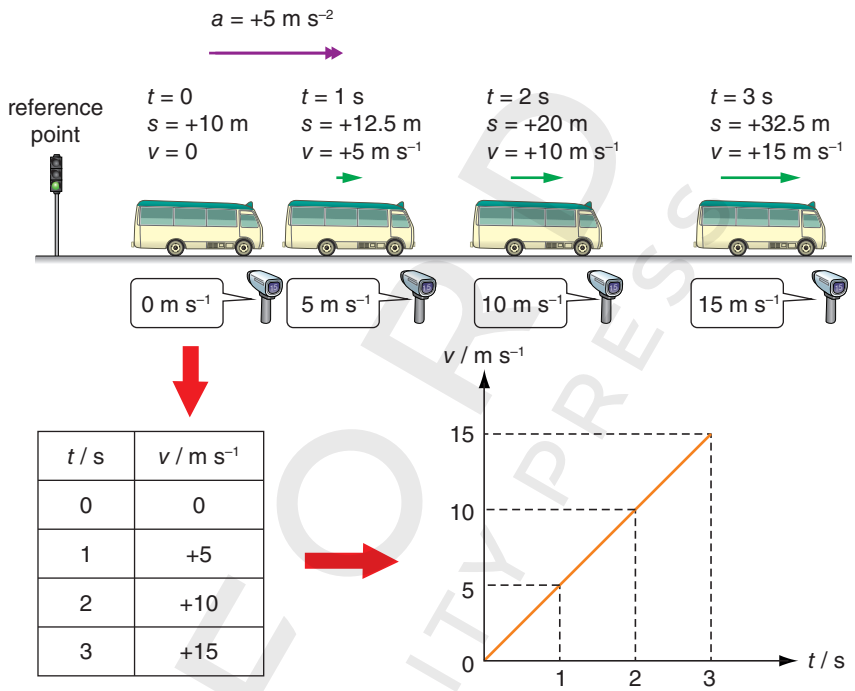
Know more

Speed gun

A speed gun is a device used to measure the speed of a moving object. It makes use of the reflection of radio wave or laser light to measure speeds.

velocity–time graph 速度—時間關係線圖

Suppose another car moves forwards from rest with a constant acceleration of +5 m s^{-2}. Since the velocity of the car increases by 5 m s^{-1} each second, its v–t graph is as follows (Fig 2.2i).



t / s	v / m s^{-1}
0	0
1	+5
2	+10
3	+15

Fig 2.2i *Plotting the v–t graph of a car in uniformly accelerated motion.*

i Slope of a *v*–*t* graph

The slope of the graph gives the acceleration of the object.

$$\text{Slope of } v\text{–}t \text{ graph} = \frac{\Delta v}{\Delta t} = \text{acceleration}$$

Unit of slope $= \frac{\text{m s}^{-1}}{\text{s}}$
$= \text{m s}^{-2}$
= unit of acceleration

From this relation, the following can be obtained from a v–t graph.

1 If the graph is a **straight line**, the object moves with a **uniform acceleration**.

2 A **steeper slope** represents an **acceleration with a larger magnitude** (Fig 2.2j). **Zero slope** means zero acceleration, i.e. **uniform motion**.

3 The **sign of the slope** gives the **direction of acceleration** (Fig 2.2k).

Ss need not calculate the acceleration from a v–t graph which is a curve. However, they should know how the acceleration changes from the slope of the curve.

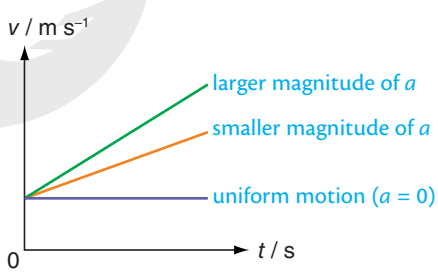


Fig 2.2j *A steeper slope represents a larger magnitude of acceleration.*

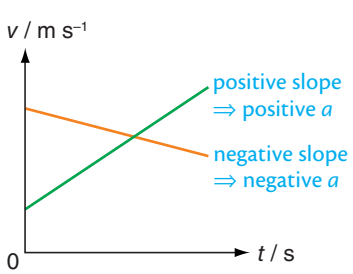
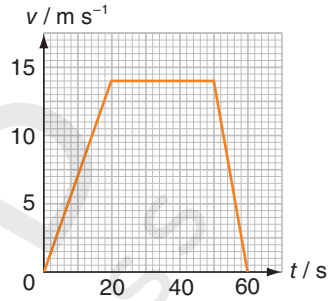


Fig 2.2k *Different signs of slope represent different direction of acceleration.*

Example 4 Finding acceleration from *v*–*t* graph

An MTR train travels along a straight track from Mong Kok to Prince Edward. Its *v*–*t* graph is as shown.



(a) Describe the motion of the train.

(b) Find the acceleration of the train at t = 10 s and 55 s.

Solution

State clearly if the acceleration is constant. We usually do not need to state the direction of motion if it remains unchanged. ▶

(a) During 0–20 s, it speeds up from rest with a constant acceleration.
During 20–50 s, it is in uniform motion.
During 50–60 s, it slows down to rest with a constant deceleration.

(b) Apply acceleration = slope of *v*–*t* graph.

At t = 10 s, acceleration = $\frac{14 - 0}{20 - 0}$ = 0.7 m s^{-2}

At t = 55 s, acceleration = $\frac{0 - 14}{60 - 50}$ = −1.4 m s^{-2}

Simulation

→ This simulation allows Ss to freely draw any *v*–*t* graph and observe the corresponding motion.

To clarify, the following can be obtained from the velocities read directly from the *v*–*t* graph, but not from the slope of the graph (Fig 2.2l).

1 The larger the magnitude of v, the higher the speed of the object.

2 The object changes its direction of motion when the sign of its velocity changes, i.e. when the graph crosses the horizontal axis.

3 The object is at rest when its velocity is zero.

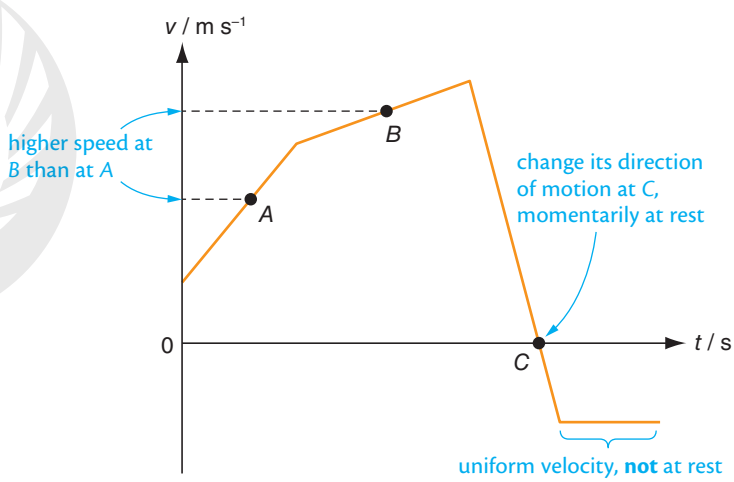


Fig 2.2l *Information obtained by reading the velocities directly.*

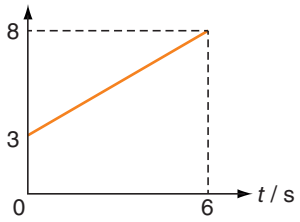
Google Forms Microsoft Forms

Quick check 5

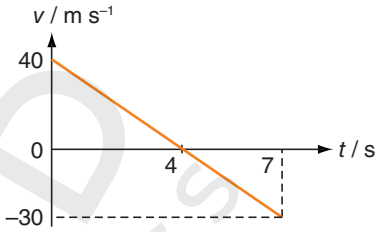
→ E-version in OUP Exercise Platform is also available.

1 Find the acceleration of the object in each of the following v–t graphs.

(a)



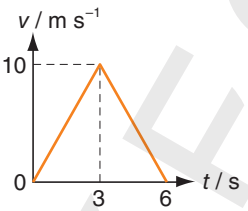
(b)



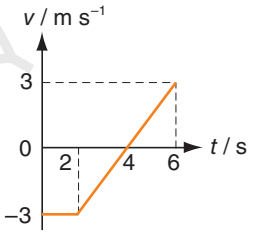
Learn by practice 3

1 The following v–t graphs show the motion of different objects moving along a straight line. Describe the motion of each object.

(a)



(b)



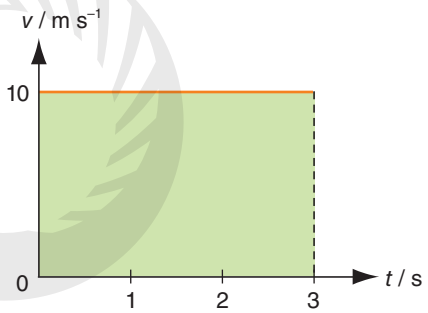
ii Area under a v–t graph

The area under a v–t graph is a product of velocity and time. In fact,

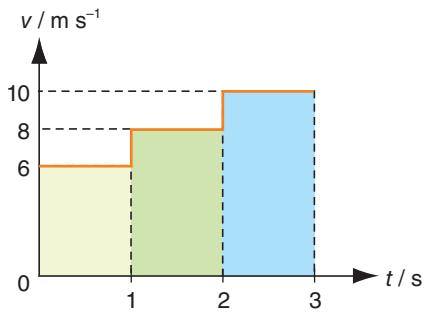
area under v–t graph = total displacement

Unit of area = m s^{-1} × s
= m
= unit of displacement

Figure 2.2m illustrates the above concept.



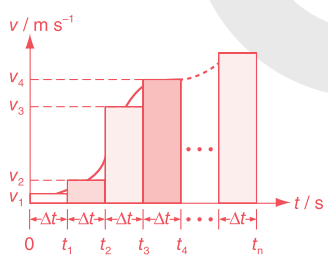
Displacement = velocity × time
= (10 m s^{-1}) × (3 s)
= 30 m
= area under graph



Displacement = displacement from 0–1 s
+ displacement from 1–2 s
+ displacement from 2–3 s
= 6 × 1 + 8 × 1 + 10 × 1
= 24 m

Fig 2.2m *The area under a v–t graph is the displacement.*

Ts may explain the following to more able Ss. Suppose the velocity changes continuously with time (v–t graph being a curve). The area under the graph can be found by summing the areas of a series of extremely thin rectangle strips as shown.



The area under the v–t graph is the displacement even when the graph is a curve.

The area under a graph can be taken as positive and negative. The **area above the time axis is taken as positive**, and the **area below the time axis is taken as negative**. The sign of the area gives the **direction of displacement**.

For example, in Figure 2.2n, a girl travels to the right from $t = 0$ to $t = 2$ s and then travels to the left from $t = 2$ s to $t = 6$ s. Her v–t graph is drawn (taking rightwards as positive) and her total displacement can be calculated from the graph as shown.

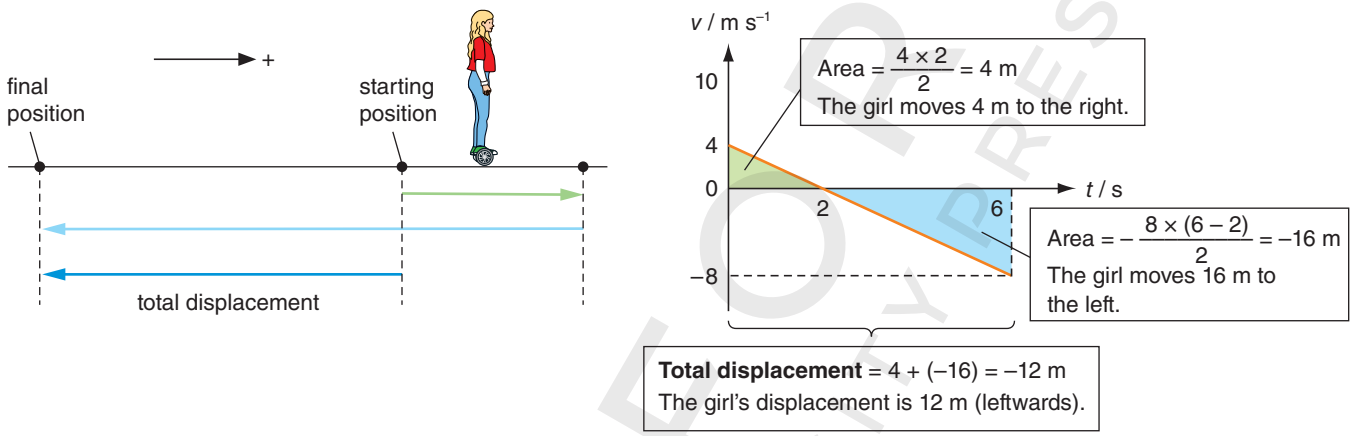


Fig 2.2n *A girl travelling along a straight road.*

Example 5 Trolley moving along an inclined plane

A trolley is given a sharp push up a smooth inclined plane from point X at time $t = 0$ (Fig a). After reaching the highest point, it slides down the plane. It moves along a straight line during the whole journey and reaches point Y at $t = 5$ s. Its v–t graph is a straight line (Fig b). Take the direction up the plane as positive.

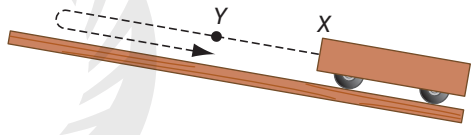


Fig a

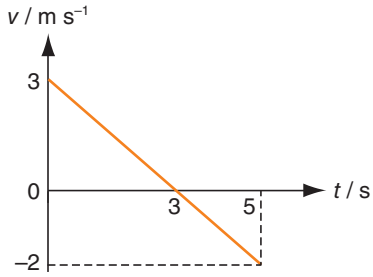


Fig b

(a) When does the trolley reach the highest point?

(b) Find the acceleration of the trolley at $t = 3$ s.

(c) Find the distance between X and Y.

(d) Find the distance travelled by the trolley in the whole journey.

Tip

When the trolley reaches the highest point, it changes its direction of motion. This happens when the v–t graph crosses the horizontal axis.

Solution

(a) At $t = 3$ s

(b) Acceleration

= slope of v–t graph

$$= \frac{0 - 3}{3 - 0}$$

$$= -1 \text{ m s}^{-2}$$

(c) Displacement over the whole journey

= total area under v–t graph

$$= \frac{3 \times 3}{2} - \frac{(5 - 3) \times 2}{2}$$

$$= 2.5 \text{ m}$$

$\therefore$ The distance between X and Y is 2.5 m.

(d) Distance travelled $= \frac{3 \times 3}{2} + \frac{(5 - 3) \times 2}{2} = 6.5$ m

Tip

This is equal to the difference between the distances travelled in the upward journey and in the downward journey.

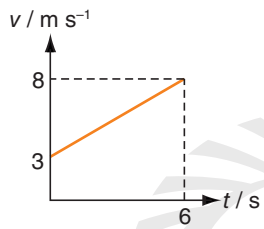
Google Forms Microsoft Forms

Quick check 6

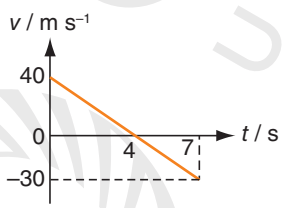
→ E-version in OUP Exercise Platform is also available.

1 Find the displacement of the object during the given time period in each of the following v–t graphs.

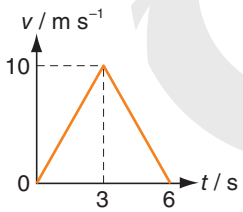
(a)



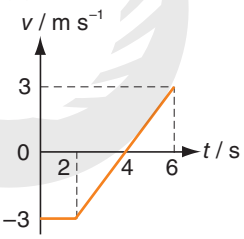
(b)



(c)



(d)



2 When is the object's displacement from the initial position the maximum in each case in Q1?

3 Two cars, A and B, start from the same position at time $t = 0$ and their v–t graphs are shown below (Fig a).

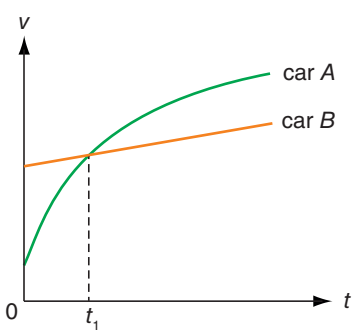


Fig a

Which of the following statements about t_1 is/are correct? Put a tick in the corresponding box(es).

☐ Cars A and B have the same speed at t_1.

☐ Cars A and B have moved the same distance from 0 to t_1.

☆ HKDSE 2015 Paper 1A Q9

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Distance between two runners

Alan and Bruce participate in a 60-m race along a straight track. They start from the same position and their v–t graphs for the first 6 s are as shown (Fig a).

What is the maximum distance between Alan and Bruce in the first 6 s?

A 3 m B 5 m

C 7.5 m D 12.5 m

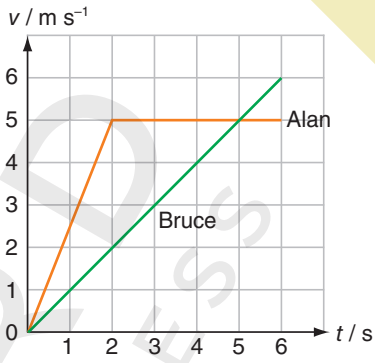


Fig a

Solution

Alan is ahead of Bruce and they become further and further apart when Alan's speed is higher than Bruce's. Therefore, the distance between them is the largest at $t = 5$ s.

Common mistake ⚠
Students may wrongly think that their distance is the largest when the difference between their speeds is the maximum, i.e. at $t = 2$ s.

Maximum distance

= distance travelled by Alan – distance travelled by Bruce

$$= \frac{(3 + 5) \times 5}{2} - \frac{5 \times 5}{2}$$

We may divide the motions into different stages which are separated by
- the moment when the direction of motion changes
- one's speed starts to exceed the other's

= area under orange line – area under green line

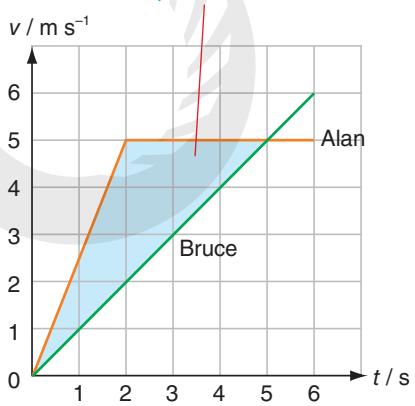
$= 7.5$ m

∴ The answer is C.

▶ Revision 2 Q14 (p.86)

In the above example, the **difference in the displacement** of Alan and Bruce in the first 5 s is actually **equal to the area bounded between the two graphs**. By considering the relative positions of the graphs, we can find out by how much a distance a person leads (Fig 2.2o).

From $t = 0$ to 5 s, Alan runs a longer distance.
Area bounded between graphs $= \frac{3 \times 5}{2} = 7.5$ m
∴ Alan leads by 7.5 m in the first 5 s.



From $t = 5$ s to 6 s, Bruce runs a longer distance.
Area bounded between graphs $= \frac{1 \times 1}{2} = 0.5$ m
∴ Alan leads by 7 m (= 7.5 – 0.5) in the first 6 s.

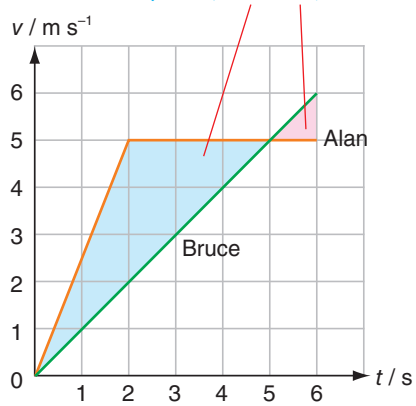


Fig 2.2o *Calculating the distance led by a person.*

c Acceleration–time graph (*a*–*t* graph)

An **acceleration–time graph** (*a*–*t* graph) is a graph that shows the acceleration of an object at different instants. The *a*–*t* graphs of a car in uniform motion and another car in uniformly accelerated motion are as shown in Figures 2.2p and 2.2q respectively.

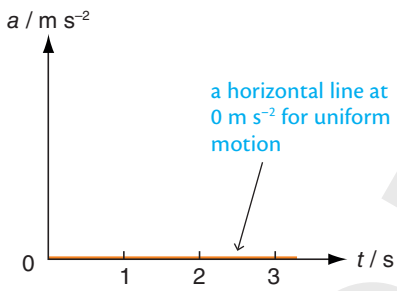


Fig 2.2p *a*–*t* graph for uniform motion.

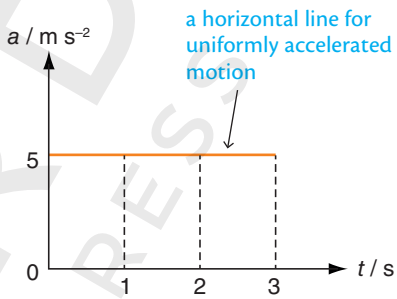


Fig 2.2q *a*–*t* graph for uniformly accelerated motion

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Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 An object accelerates from rest and then moves along a straight line. Its *a*–*t* graph is as shown (Fig a).

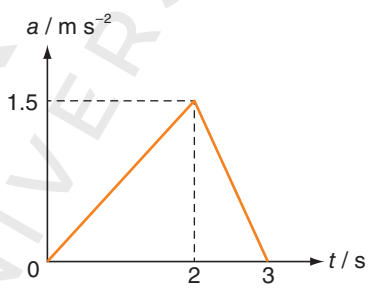


Fig a

Which of the following is/are correct?

- ☐ The object keeps accelerating in the same direction.
- ☐ The total displacement of the object is 0 from $t = 0$ to $t = 3$ s.
- ☐ The object changes its direction of motion at $t = 2$ s.

d Relationship between different motion graphs

The *s*–*t* graph, *v*–*t* graph and *a*–*t* graph have the following relationships:

$$\textit{s}\text{–}\textit{t} \text{ graph} \underset{\text{area}}{\overset{\text{slope}}{\rightleftarrows}} \textit{v}\text{–}\textit{t} \text{ graph} \xrightarrow{\text{slope}} \textit{a}\text{–}\textit{t} \text{ graph}$$

Simulation

→ This simulation shows the *s*–*t*, *v*–*t* and *a*–*t* graphs of different motions. It is available on OUP web in PC version (.exe) only.

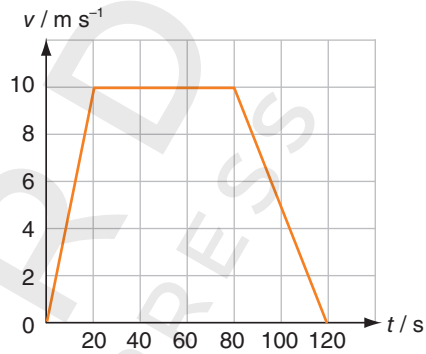
acceleration–time graph 加速度—時間關係線圖

→ In the following website, motion graphs are generated and presented with animation according to the input variables.

Example 6 Relationship between motion graphs

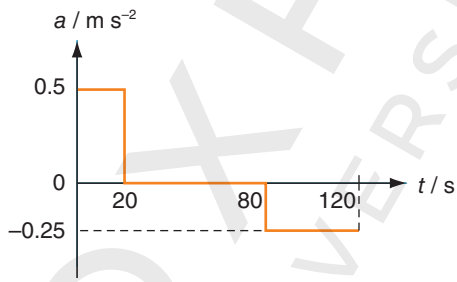
A cyclist travels forwards along a straight line from point X. He arrives at point Y at t = 120 s. His v–t graph is as shown.





Sketch the acceleration–time graph for the cyclist.

Solution



Tip

From 0–20 s,

$a = \frac{10 - 0}{20 - 0} = 0.5 \text{ m s}^{-2}$

From 20–80 s,

$a = 0$

From 80–120 s,

$a = \frac{0 - 10}{120 - 80} = -0.25 \text{ m s}^{-2}$

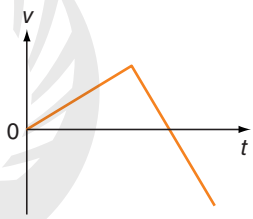
Google Forms Microsoft Forms

Quick check 8

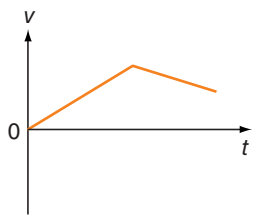
→ E-version in OUP Exercise Platform is also available.

1 Match the v–t graphs A–C with the corresponding a–t graphs I–III.

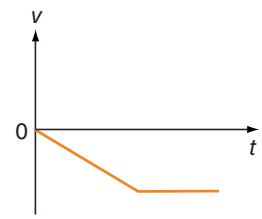
A



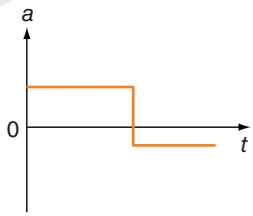
B



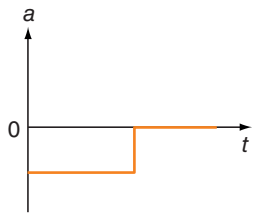
C



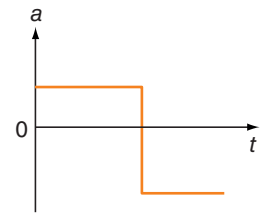
I



II



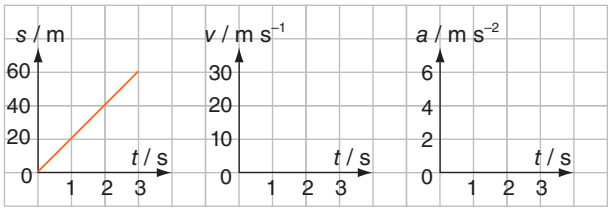
III



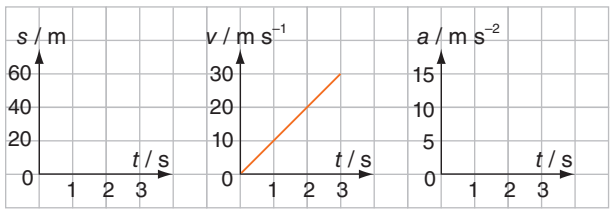
Learn by practice 4

1 Complete the motion graphs for the cars.

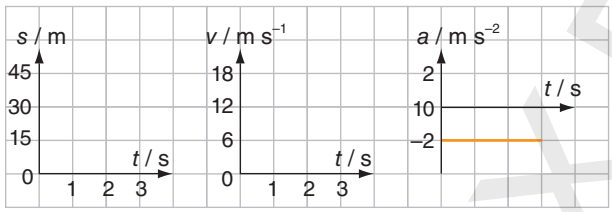
Car *A* (given *s*–*t* graph)



Car *B* starting at *s* = 0 (given *v*–*t* graph)

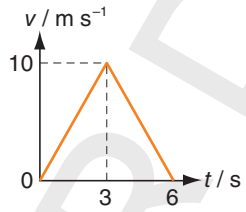


Car *C* with an initial velocity of 18 m s^{-1}, starting at *s* = 0 (given *a*–*t* graph)

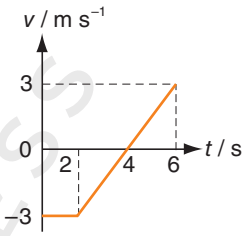


2 The following *v*–*t* graphs show the motion of different objects moving along a straight line. Sketch the *a*–*t* graph for each object.

(a)



(b)



3 The following *a*–*t* graph shows the motion of an object moving along a straight line (Fig a). The object is at rest at *t* = 0. Describe its motion from *t* = 0 to *t* = 10 s.

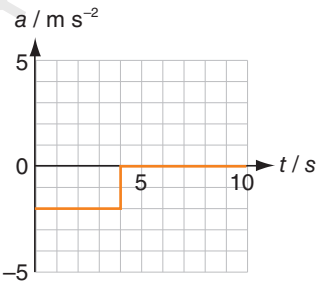


Fig a

2 Motion analysing tools

In laboratories, we can produce motion graphs conveniently with a ***motion sensor*** (Fig 2.2r). It can be connected to a mobile device or computer that runs a data-logging app or program. The *s*–*t*, *v*–*t* and *a*–*t* graphs of an object moving towards or away from the sensor can be produced (Fig 2.2s).



Fig 2.2r *A wireless motion sensor.*

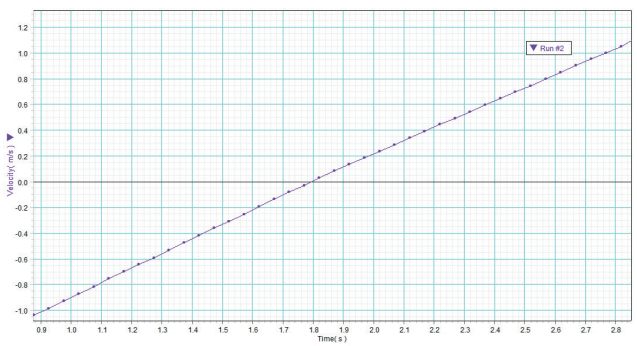


Fig 2.2s *A v–t graph produced by a data-logging program. (A trolley being given a sharp push moves up an inclined runway and then slides down.)*

motion sensor 運動感應器

→ Teaching notes and exercises are available on OUP web.

Skill
Experiment

Cross-link
You will learn more about ultrasound in Book 3B Ch 7. ▶

Motion sensors

A motion sensor can be used to detect an object moving along a straight line towards or away from it. It emits *ultrasound* signals and detects echoes from the object repeatedly. From the time lapse between sending and receiving the signals, the positions of the object at different instants can be calculated. The results can be shown in the form of motion graphs.



Fig 2.2t *A trolley with built-in sensors.*

Some trolleys (Fig 2.2t) are equipped with built-in sensors. They can be connected to mobile devices or computers via a wireless network. The built-in sensors can record the trolley's displacement, velocity, acceleration, etc. The results can also be shown in the form of motion graphs.

Some software can help you plot the successive positions of an object in motion from a recorded video. The motion graphs can then be generated instantly.

→ This video demonstrates Expt 2a.

Video Simulation

→ The simulation 'Studying motion using a motion sensor' simulates step 1 of Expt 2a.

→ The simulation 'Acceleration down a slope' simulates step 2 of Expt 2a.

Experiment 2a Using motion analysing tools

1 Set up a motion sensor (Fig a). Start data-logging. Walk towards and away from the sensor along a straight line. Note the s–t and v–t graphs obtained.

2 Put a trolley on a gently inclined runway and put the motion sensor at the upper end of the runway (Fig b). Start data-logging and release the trolley. Note the s–t and v–t graphs obtained.

3 Release a trolley on the runway again and record a video of its motion from sideways. Import the video to the motion tracking software. Note the s–t and v–t graphs obtained.

You may also use a trolley with built-in sensor in this step. ▶

Keep a safe distance from the runway and trolleys.

Remind Ss that the motion sensor may capture false signals if something gets near to the target object.

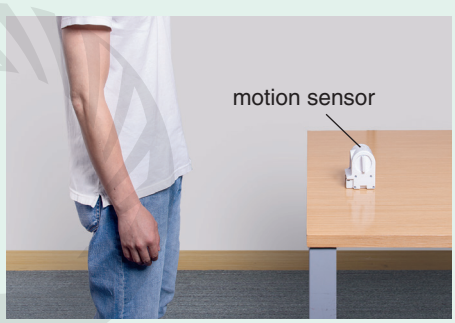


Fig a

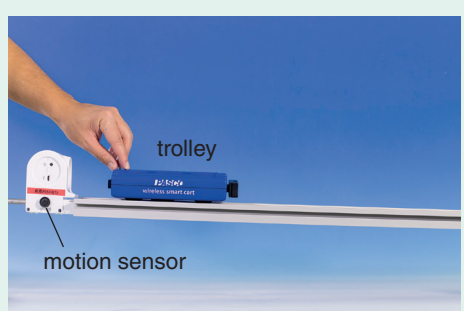


Fig b

Discussion

What do the graphs tell us about the motion?

ultrasound 超聲波

Exercise 2.2

1 The v–t graph of a car is shown in Figure a.

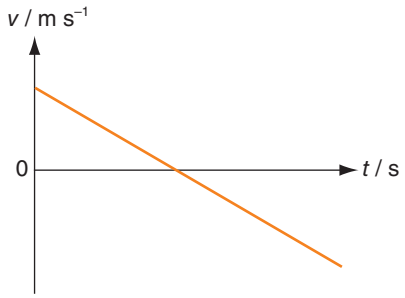


Fig a

Oil in the car steadily drips out and leaves some dirt marks on the road. Which of the following dirt mark patterns is correct?

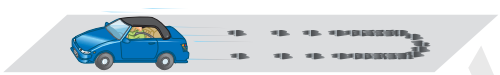
A



B



C



D



***2** Figure b shows the s–t graph of a car moving along a straight road.

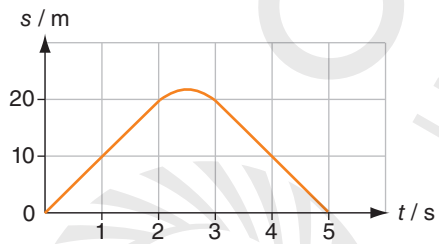


Fig b

Which of the following statements about the motion of the car is/are correct?

(1) From $t = 0$ to $t = 2$ s, it moves at a constant velocity of 10 m s^{-1}.

(2) From $t = 2$ s to $t = 3$ s, it slows down, stops momentarily and then speeds up in the opposite direction.

(3) From $t = 2$ s to $t = 3$ s, its average acceleration is −10 m s^{-2}.

A (2) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

***3** Figure c shows the v–t graph of an object moving along a straight line.

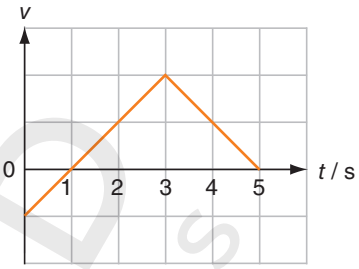


Fig c

Which of the following statements about its motion is/are correct?

(1) Its acceleration changes from negative to positive at $t = 1$ s.

(2) Its moving direction changes at $t = 3$ s.

(3) It is momentarily at rest at $t = 1$ s.

A (2) only B (3) only

C (1) and (2) only D (1) and (3) only

***4** A squirrel moves along a straight line on a tree (Fig d). Its displacement s from the trunk changes with time t as shown (Fig e).



Fig d

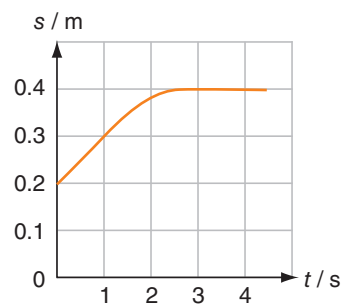


Fig e

Which of the following statements about the motion of the squirrel is/are correct?

(1) Its total displacement is 0.4 m from $t = 0$ to $t = 4$ s.

(2) Its velocity is 0.3 m s^{-1} at $t = 1$ s.

(3) It is stationary at $t = 3.5$ s.

A (2) only B (3) only

C (1) and (3) only D (1), (2) and (3)

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★**5** Figure f shows the a–t graph of an object moving along a straight line.

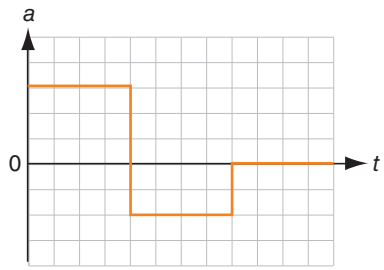
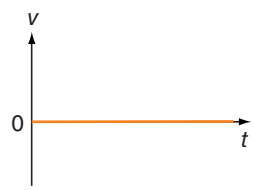
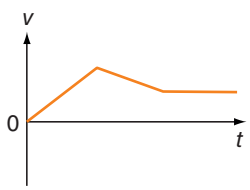
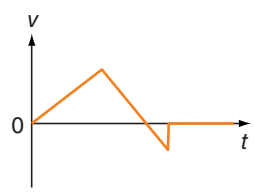


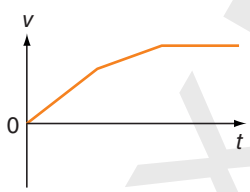
Fig f

Which of the following is possibly the v–t graph of the object?

A


B


C


D


★**6** Figure g shows the positions of an object at different instants.

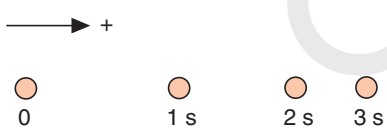
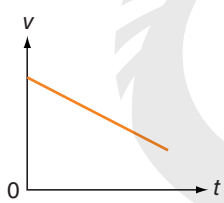
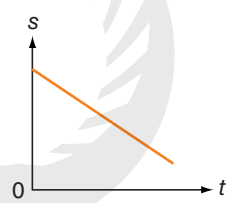
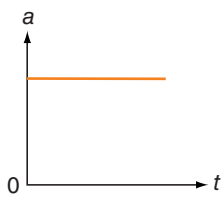


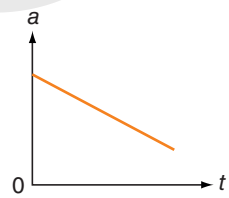
Fig g

Which of the following graphs best describes the motion of the object?

A


B


C


D


7 Figure h shows the s–t graphs of objects P and Q. Figure i shows the v–t graphs of objects X and Y.

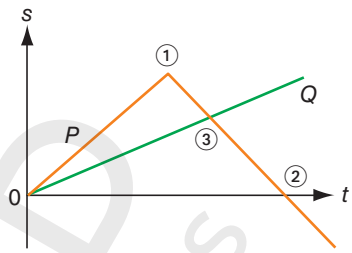


Fig h

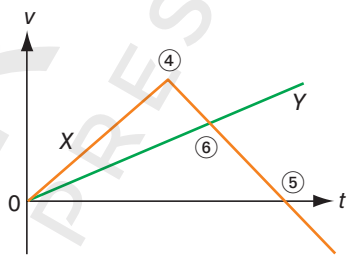


Fig i

Which point(s) in the graphs indicate(s)

(a) an object reverses its moving direction?

(b) an object returns to its starting position?

(c) the two objects meet?

★**8** Thomas rides a bicycle up a slope along a straight road. He leaves point X at time $t = 0$ and arrives at point Y at $t = 30$ s. Figure j shows his v–t graph. The direction up the slope is taken as positive.

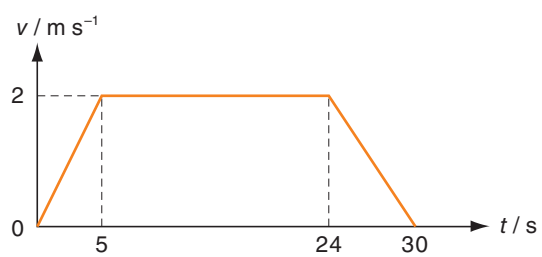


Fig j

(a) Describe his motion from $t = 0$ to $t = 30$ s.

(b) Sketch his a–t graph from $t = 0$ to $t = 30$ s.

(c) Find his total displacement from $t = 0$ to $t = 30$ s.

(d) He stays at Y from $t = 30$ s to $t = 36$ s. Then he moves down the slope at a constant acceleration. His speed is 7 m s^{-1} when he returns to X at $t = T$.

(i) Sketch the v–t graph of Thomas from $t = 0$ to $t = T$.

(ii) Find T.

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★**9** Two objects X and Y move along a straight line. Figure k shows their s–t graphs.

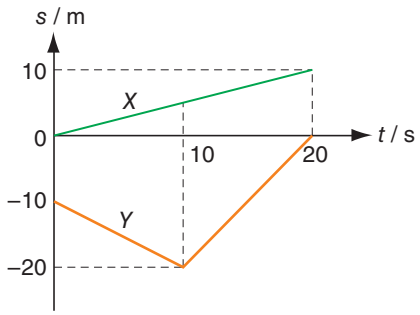


Fig k

(a) What is the largest distance between the two objects from $t = 0$ to $t = 20$ s?

(b) Find their average velocities from $t = 0$ to $t = 20$ s.

(c) Find their instantaneous velocities at $t = 5$ s and $t = 15$ s.

★**10** An object initially moves at a velocity u. After speeding up with a uniform acceleration a for time t, its velocity becomes v while its displacement is s.

(a) Sketch the object's velocity–time graph.

(b) By considering the slope of the graph in (a), show that $v = u + at$.

(c) By considering the area under the graph in (a), show that $s = \frac{(u + v)t}{2}$.

★**11** X and Y run along a straight line. They start from the same position and their v–t graphs are as shown (Fig l).

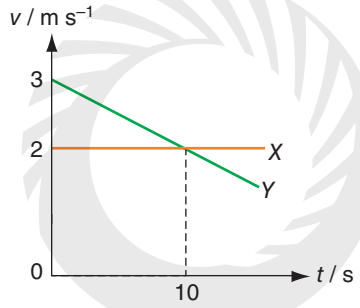


Fig l

(a) Describe the motion of X and Y from $t = 0$ to $t = 10$ s.

(b) Find the maximum distance between X and Y in the first 10 s.

★**12** In the story *The Tortoise and the Hare*, the tortoise wins the race because the hare sleeps midway through the race. Suppose they participate in a 60-m race and their s–t graphs are as shown (Fig m).

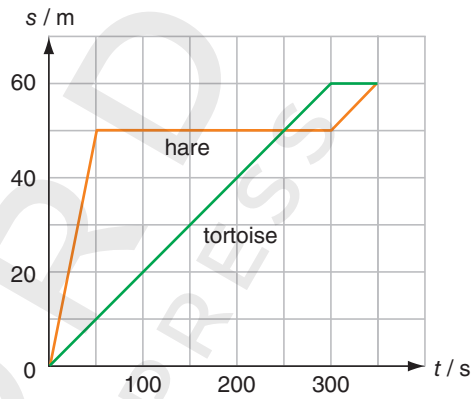


Fig m

(a) When does the tortoise catch up with the hare?

(b) What is the biggest distance led by the hare in the race?

(c) Find their highest velocities in the race.

(d) Find the average velocity of the hare in the race.

★**13** Figure n shows the v–t graphs of two cars A and B. They travel along a straight road from the same position.

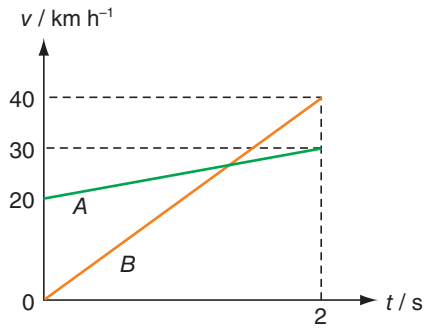


Fig n

(a) Find the acceleration of A in the first 2 s.

(b) Which car takes the lead at $t = 2$ s? What is the distance led by the car?

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2.3 Equations of uniformly accelerated motion

There are skid marks on the road in some traffic accidents. These marks are produced when hard braking is applied to make the wheels stop rolling, making the tires slide on the road. The skid marks can help the police to estimate the initial speed of the car (Fig 2.3a). It can be done by using some equations, which we will learn in the following part.

Flipped classroom
Watch this and answer the questions.

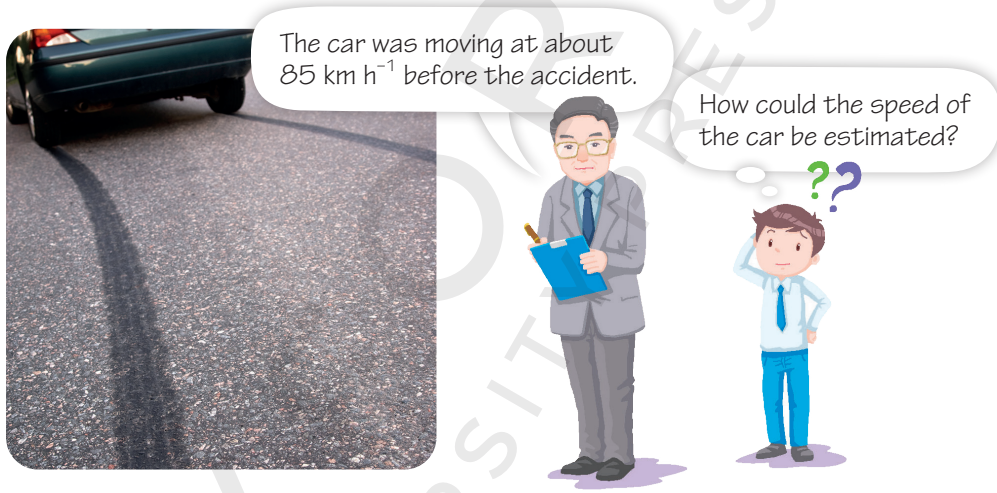


Fig 2.3a Estimating the initial speed of car from its skid marks.

1 Deriving the equations of motion

DSE goal
Derive equations of uniformly accelerated motion.

The motion of the car during skidding can be considered as a uniformly accelerated motion, which refers to motion along a straight line with a constant acceleration (***a* = constant**). Consider an object moving with uniform acceleration a (Fig 2.3b). Its initial velocity is u. After accelerating for time t, its final velocity is v. Its displacement within this period of time is s. Its velocity–time graph is a straight line as its slope (= a) is constant (Fig 2.3c).

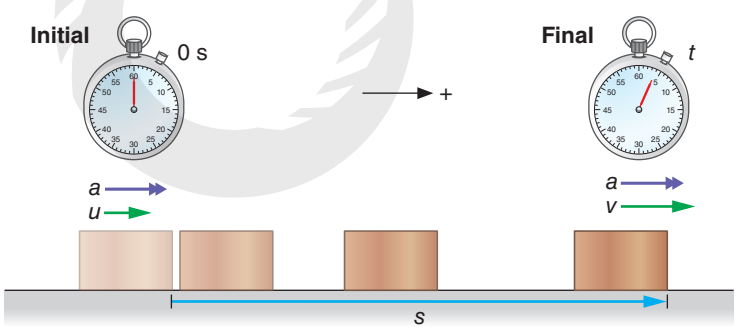


Fig 2.3b An object accelerates uniformly from u to v along a straight line in time t.

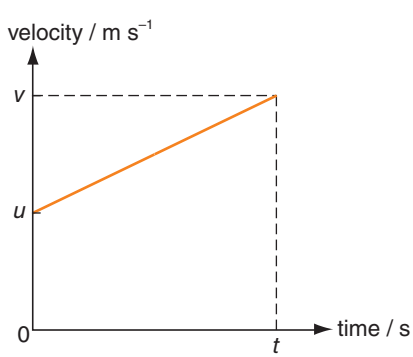


Fig 2.3c The v–t graph of an object which accelerates uniformly.

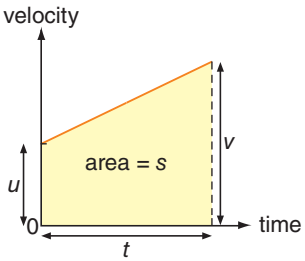
Now we can derive four important equations for uniformly accelerated motion.

By the definition of acceleration,

$$a = \frac{v - u}{t}$$

$$v - u = at$$

$$v = u + at \quad \text{.....(1)}$$



▶ The area under the velocity–time graph (Fig 2.3c on p.62) is equal to the displacement:

$$s = \frac{1}{2}(u + v)t \quad \text{.....(2)}$$

Substitute (1) into (2):

$$s = \frac{1}{2}(u + u + at)t$$

$$s = \frac{1}{2}(2ut + at^2)$$

$$s = ut + \frac{1}{2}at^2 \quad \text{.....(3)}$$

Rearrange $a = \frac{v - u}{t}$ as $t = \frac{v - u}{a}$ and substitute it into (2):

$$s = \frac{1}{2} \times (u + v) \times \frac{v - u}{a}$$

$$2as = (v + u) \times (v - u)$$

▶ $2as = v^2 - u^2$

$$v^2 = u^2 + 2as \quad \text{.....(4)}$$

$(v + u) \times (v - u)$
$= v^2 - vu + uv - u^2$
$= v^2 - u^2$

Try Exercise 2.3 Q7 (p.70) to derive the equation $s = vt - \frac{1}{2}at^2$.

For an object moving along a straight line with uniform acceleration,

$$\boldsymbol{v = u + at} \quad \textbf{.....(1)}$$

$$\boldsymbol{s = \frac{(u + v)t}{2}} \quad \textbf{.....(2)}$$

$$\boldsymbol{s = ut + \frac{1}{2}at^2} \quad \textbf{.....(3)}$$

$$\boldsymbol{v^2 = u^2 + 2as} \quad \textbf{.....(4)}$$

u: initial velocity
v: final velocity
s: displacement
a: acceleration
t: time

These four equations are called the **equations of uniformly accelerated motion**.

equations of uniformly accelerated motion 勻加速運動方程

DSE goal

Solve problems involving objects in uniformly accelerated motion.

DSE exam

14(1A)Q5, 16(1A)Q7, 17(1A)Q6

2 Applying the equations of motion

Note the following in applying the equations of motion to solve problems.

1 These equations apply to uniformly accelerated motion only (***a* = constant**).

2 If we know **any three among the quantities *a, u, v, s* and *t***, the remaining quantities can be found using these equations.

3 The signs of s, u, v and a should be consistent with the positive direction defined for the problem.

Let us see how these equations can be used to solve problems.

Example 7 Braking a snowmobile

A snowmobile moves along a straight line. At a certain instant, it starts braking to decelerate uniformly. Its speed decreases from 25 m s^{-1} to 15 m s^{-1} after travelling 40 m.

(a) Find the acceleration of the snowmobile.

(b) How much time does the snowmobile take to decelerate from 15 m s^{-1} to rest?



Solution

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Applying equations of motion

① Define the positive direction.

② Find out which variables are known, and which are to be found. The signs of the variables should be consistent with the positive direction. Remember to use proper units for the variables.

③ Choose the suitable equation. Each equation should have 3 known variables and only 1 unknown.

(a) Take forwards as positive.

$u = +25$ m s^{-1}, $v = +15$ m s^{-1}, $s = +40$ m, $a = ?$

By $v^2 = u^2 + 2as$,

$15^2 = 25^2 + 2a \times 40$

$a = -5$ m s^{-2}

The acceleration of the snowmobile is 5 m s^{-2} (backwards).

(b) Consider the motion from 15 m s^{-1} to rest.

$u = +15$ m s^{-1}, $v = 0$, a $= -5$ m s^{-2}, $t = ?$

By $v = u + at$,

$0 = 15 + (-5)t$

$t = 3$ s

It takes 3 s to decelerate from 15 m s^{-1} to rest.

Example 8 Skid marks

This example shows how the skid marks can help the police to estimate the initial speed of the car as asked on p.62. ▶

A police officer is investigating a car accident. The car stopped after leaving skid marks of 20 m. It is also found from CCTV that the braking lasted 1.7 s. What was the speed of the car just before it braked?

Solution

Take forwards as positive.

$s = +20$ m, $t = 1.7$ s, $v = 0$, $u = ?$

By $s = \frac{(u + v)t}{2}$,

$20 = \frac{(u + 0) \times 1.7}{2}$

$u = 23.5$ m s^{-1}

The speed of the car was 23.5 m s^{-1} just before it braked.

Example 9 Stopping distance

A car travels at 63 km h^{-1} along a straight road. The driver sees an obstacle 30 m ahead and starts braking the car after a reaction time of 0.8 s. The car slows down uniformly at 7 m s^{-2} during braking.

(a) Find the thinking distance, i.e. the distance travelled during the driver's reaction time.

(b) Find the braking distance, i.e. the distance travelled during braking to a stop.

(c) Does the car hit the obstacle?

Solution

(a) 63 km h^{-1} $= \frac{63}{3.6}$ m s^{-1} $= 17.5$ m s^{-1}

Thinking distance = speed × reaction time = $17.5 \times 0.8 = 14$ m

✎ Assume the car moves at a constant velocity during the reaction time.

(b) Consider the motion during braking.

Take forwards as positive.

$u = +17.5$ m s^{-1}, $v = 0$, $a = -7$ m s^{-2}, $s = ?$

By $v^2 = u^2 + 2as$,

$0^2 = 17.5^2 + 2(-7) \times s$

$s = 21.9$ m

The braking distance is 21.9 m.

Tip

The stopping distance, i.e. the total distance for the car to stop after the driver notices the obstacle, is equal to 35.9 m (= 14 + 21.9), which is longer than 30 m. ▶

(c) Yes

STSE

Safe gaps between cars

The stopping distance of a car is the sum of the thinking distance and braking distance. Travelling at a higher speed needs a much longer stopping distance (Fig a). It can be illustrated by using v–t graphs (Fig b).

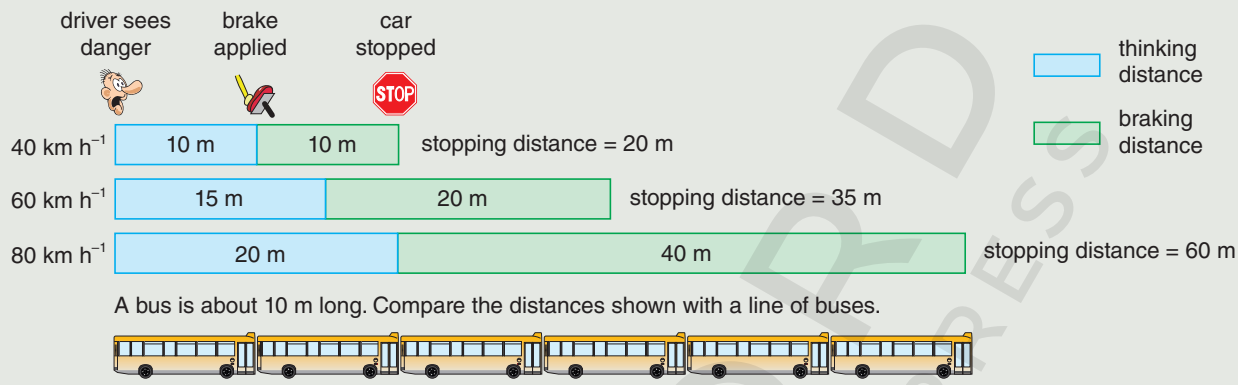


Fig a

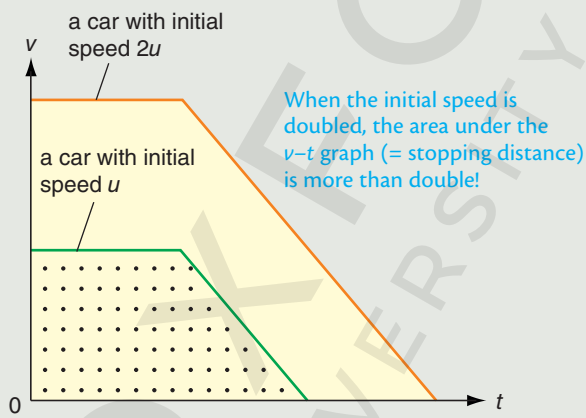


Fig b

Drivers are told to maintain a safe gap from the car in front. A commonly used safety rule is to leave a 'two-second gap'—a car should pass a point at least 2 s after the car in front passes it. The length of this gap will increase with the speed of the cars. This ensures the gap is large enough for the car at the back to stop when accident happens to the car in the front.

Besides speed, some other factors can also affect the stopping distance. For example, if the road is wet, the deceleration of a car will be lower and so the stopping distance will be larger. Can you think of other factors?

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Apply the equations of uniformly accelerated motion to complete the table.

	u / m s^{-1}	v / m s^{-1}	s / m	a / m s^{-2}	t / s	Equation used
(a)	30	–	8		2	
(b)	1	3	–	2		
(c)		–2	10.5	–1	–	
(d)	4	–2		–	5	

Learn by practice 5

1 A car speeds up uniformly from rest to a speed of 25 m s^{-1}. If the process takes 4 s, find its acceleration.

2 An athlete starts from rest with a uniform acceleration of 3 m s^{-2} along a straight track. How long does he take to run the first 6 m?

3 Peter runs on a straight road with an initial speed of 5 m s^{-1}. He speeds up with a uniform acceleration of 3 m s^{-2}. Find the total distance travelled by him just before his speed reaches 13 m s^{-1}.

4 A car travelling at 72 km h^{-1} on a highway tries to speed up with a uniform acceleration of 0.1 m s^{-2}. Find the time needed for the car to speed up to 100 km h^{-1}.

5 An apple fell from rest with an acceleration of 9.81 m s^{-2} and hit Isaac's head after 0.4 s (Fig a). Find the initial height of the apple above Isaac's head.

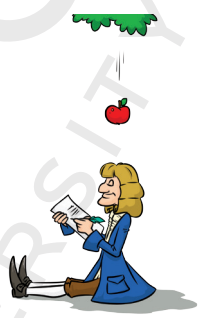


Fig a

6 A bullet is fired towards a wooden wall which is 0.15 m thick (Fig b).

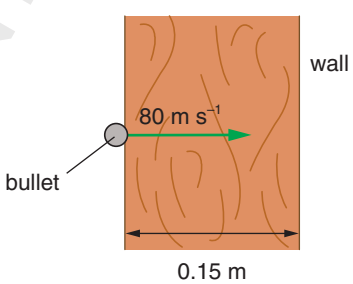


Fig b

The bullet penetrates the wall in 2.5×10^{-3} s. If the speed of the bullet is 80 m s^{-1} when it enters the wall, what is its speed when it leaves the wall? Assume that the bullet moves along a straight line in the whole process and its deceleration inside the wall is uniform.

7 A car travelling at 30 m s^{-1} along a straight road stops in 3 s after the driver sees an obstacle. The driver has a reaction time of 0.8 s.

(a) Find the thinking distance.

(b) The deceleration of the car is uniform. Find the braking distance.

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3 Equations of motion and motion graphs

For an object moving with a given initial velocity u and uniform acceleration a, the following graphs can be obtained by considering the equations of motion (Table 2.3a).

Table 2.3b shows some graphs corresponding to motion starting from rest ($u = 0$).

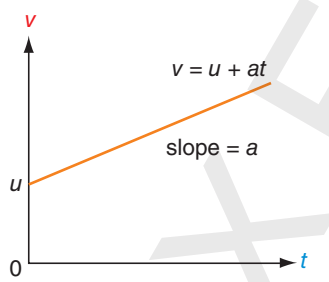
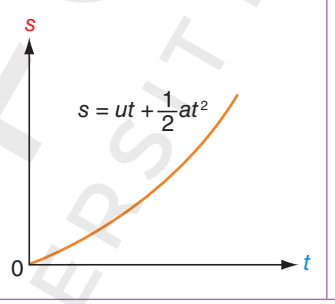
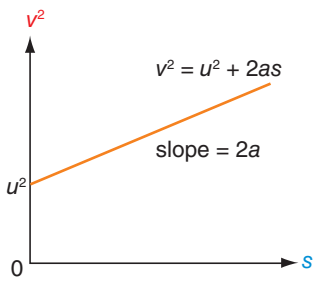
given a and u		
$v = u + at$ • v varies linearly with t ⇒ straight-line graph • slope = a • y-intercept = u	$s = ut + \frac{1}{2}at^2$ • s does not vary linearly with t ⇒ curved graph • passing through origin	$v^2 = u^2 + 2as$ • v^2 varies linearly with s ⇒ straight-line graph • slope = $2a$ • y-intercept = u^2
		

Table 2.3a *Different graphs of uniformly accelerated motion.*

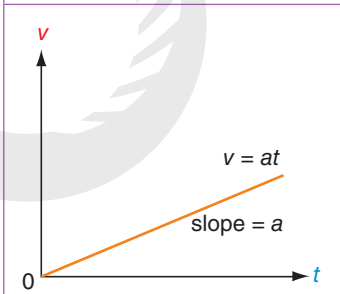
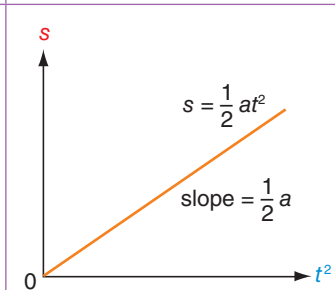
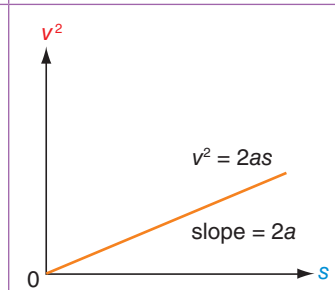
given a, $u = 0$		
$v = \cancel{u^0} + at = at$ • $v \propto t \Rightarrow v$–$t$ graph is a straight line through origin • slope = a	$s = \cancel{ut^0} + \frac{1}{2}at^2 = \frac{1}{2}at^2$ • $s \propto t^2 \Rightarrow s$–$t^2$ graph is a straight line through origin • slope = $\frac{1}{2}a$	$v^2 = \cancel{u^0} + 2as = 2as$ • $v^2 \propto s \Rightarrow v^2$–$s$ graph is a straight line through origin • slope = $2a$
		

Table 2.3b *Different graphs of uniformly accelerated motion starting from rest.*

The discussion in this part is based on one idea of mathematics: the equation of a straight line. You may visit this part after learning the mathematics. For a quick view, you may refer to the mathematics skills below.

Skill *Mathematics*

Linear relation and direct proportion

▸ p.453

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Example 10 Equations of motion and motion graphs

Susan goes down a slide and accelerates uniformly from rest along a straight line (Fig a). Her displacement from her starting position is s when the time of travel is t. Figure b shows how s varies with t^2. What is Susan's acceleration?



Fig a

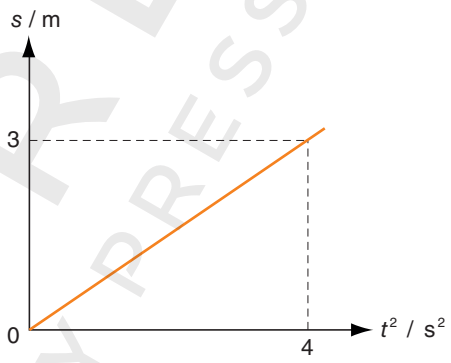


Fig b

Solution

$u = 0$ ▶

$$s = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2$$

Slope of s–t^2 graph $= \frac{1}{2}a$

$$\frac{3}{4} = \frac{1}{2}a$$

$$a = 1.5 \text{ m s}^{-2}$$

Susan's acceleration is 1.5 m s^{-2}.

Exercise 2.3

★**1** A horse accelerates uniformly from rest. Its velocity is v after accelerating for a displacement s. What is its velocity after accelerating for a total displacement $2s$?

A v

B $1.41v$

C $2v$

D $4v$



★**2** At a certain instant, cyclists X and Y are both moving towards the finish line at 10 m s^{-1} along a straight road. Cyclist X has 150 m left and is moving uniformly at 10 m s^{-1}. Cyclist Y has 250 m left and is accelerating uniformly at 0.5 m s^{-2}. Who will win the race?

A Cyclist X

B Cyclist Y

C Both will finish at the same time.

D Cannot be determined

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3 A watch dog sitting on the ground sees a thief 10 m away. It starts running towards the thief and catches him in 5 s after he has run for 20 m (Fig a).

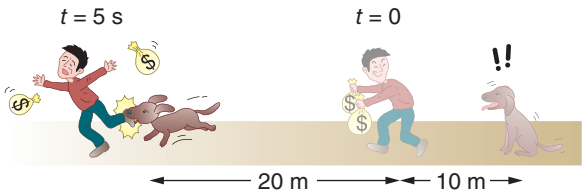


Fig a

If the dog accelerates uniformly in a straight line until catching the thief, what is its acceleration?

4 An airplane accelerates uniformly to reach a speed of 320 km h^{-1} until taking off. It is known that the airplane takes 20 s to accelerate from rest to 128 km h^{-1}.



Fig b

(a) Find the acceleration of the airplane.

(b) How far does the airplane move from rest until taking off?

5 A penguin dives into the water (Fig c). It decelerates uniformly at 6 m s^{-2} along a straight line. Its initial velocity is 5 m s^{-1}. Find its velocity after decelerating for 0.5 s.



Fig c

6 A girl on a bicycle speeds up uniformly in a straight line at 2 m s^{-2}. Her velocity is 14 m s^{-1} after 5 s. Find the girl's displacement in these 5 s.

***7** An object moves with uniform acceleration a along a straight line. Its initial velocity is u. After time t, its velocity becomes v. The displacement within this period is s.

(a) Show that $s = vt - \frac{1}{2}at^2$.

(b) Use the equation in (a) to solve Q6.

***8** Object X accelerates uniformly from rest along a straight line. Its displacement is s after time t. The graph below shows the variation of s with t^2 (Fig d).

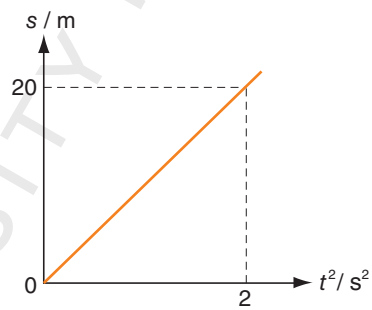


Fig d

(a) What is the acceleration of X?

(b) Object Y moves along the same path from rest with an acceleration double that of X. Sketch the s–t^2 graph for Y on Figure d.

***9** An object at point O starts to accelerate uniformly along a straight line. Its velocity is v when its displacement from O is s. Figure e shows how v^2 varies with s over the journey.

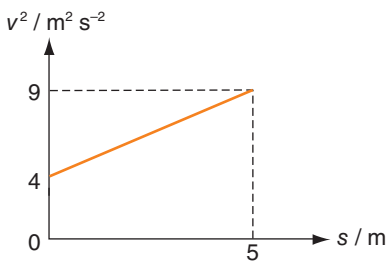


Fig e

(a) State the equation of motion that relates v^2 and s.

(b) What is the speed of the object at O?

(c) What is the acceleration of the object?

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2.4 Vertical motion under gravity

In the past, people believed that heavy objects fell faster than light objects. *Galileo Galilei* (Fig 2.4a) questioned this old belief. He was said to drop two balls of different weights at the same time from the *Tower of Pisa* (Fig 2.4b) to demonstrate that objects of different weights fell equally fast.

DSE goal

Examine the motion of free-falling objects experimentally and estimate the acceleration due to gravity.

Present graphically information on vertical motions under gravity.

Describe the effect of air resistance on the motion of objects falling under gravity.

Apply equations of uniformly accelerated motion to solve problems involving objects in vertical motion.

DSE exam

13(1A)Q8, 14(1A)Q9, 17(1B)Q4(a)(i), 19(1B)Q3(a)(i), 20(1A)Q6

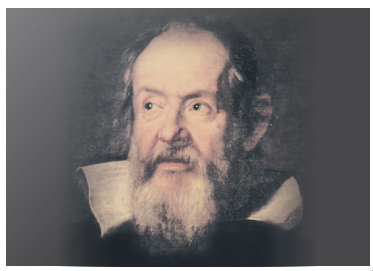


Fig 2.4a *Galileo Galilei.*



Fig 2.4b *Tower of Pisa.*

Know more

Galileo Galilei (1564-1642)

Galileo probably did not really drop the balls from the Tower of Pisa. However, he did other experiments to study falling objects.

▸ p.92

However, based on our experience in daily life, it is not difficult to find two objects falling at a different rate (such as a paper and a textbook). Let us find out the reason by doing the following experiment and activity.

After Galileo suggested that every object fell at the same rate under gravity, Boyle used his newly invented vacuum pump to remove air from a long glass tube containing a feather and a lump of lead. Boyle found that in a vacuum the feather and the lead fell together at the same rate down the tube. This experiment is often repeated in the laboratory with a coin and a feather, so it is called the 'coin and feather experiment'.

Video

→ This video demonstrates Expt 2b.

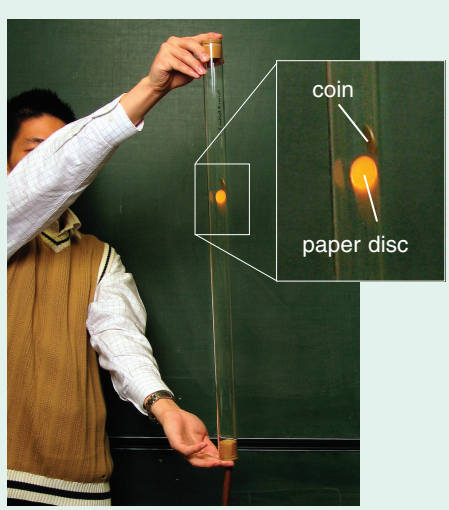
→ This is a Video Quiz about this experiment.

Experiment 2b The 'coin' and 'feather' experiment

1 Put a coin and a small paper disc (to represent the feather) in a glass tube.

2 Allow them to fall in the tube.

3 Remove air from the tube and repeat step 2.



Discussion

Does the coin always fall faster than the paper disc?

During the Apollo 15 mission, astronaut *David Scott* dropped a hammer and a feather on the Moon, an environment without air. Similar experiment was carried out in a very large vacuum chamber on the Earth.

Video

Galileo Galilei 伽利略　Tower of Pisa 比薩斜塔　David Scott 史考特

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Which one falls faster?

What you need



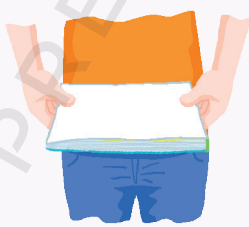
What to do

1



Drop the paper and the book at the same time.

2



Put the paper on top of the book and drop them at the same time.

What you find

Which object falls faster in each case?

Cross-link

Air resistance will be discussed in detail in Ch 4. ▶

✎ The word 'gravity' came from the Latin word for heaviness.

When you release an object from a height, it falls under **gravity**. At the same time, there is also **air resistance** obstructing the motion of the object. For objects with a similar shape, the effect of air resistance is more significant on the lighter object. This explains why the paper disc falls more slowly than the coin in the presence of air in Experiment 2b.

When the air inside the tube is removed, the effect of air resistance is eliminated and the objects **fall with the same acceleration under gravity**. Similarly, in the Hands-on activity, when you put the paper on top of the book, the air resistance acting on the paper is eliminated and therefore it falls as fast as the book.

1 Free fall

✎ Projectile motion is also a kind of free-fall motion.

When an object falls under the influence of **gravity only**, its motion is called **free fall**. All objects have the same acceleration when they fall freely near the Earth's surface. This acceleration is called **acceleration due to gravity** and is denoted by g. Let us estimate the value of g in the following experiment.

gravity 重力　air resistance 空氣阻力　free fall 自由下落　acceleration due to gravity 重力加速度

→ This video demonstrates Expt 2c.

Video Simulation

→ This simulation simulates Expt 2c.

Experiment 2c / Estimating the acceleration of free fall

1 Set up the apparatus as shown. Set the motion sensor to face downwards.

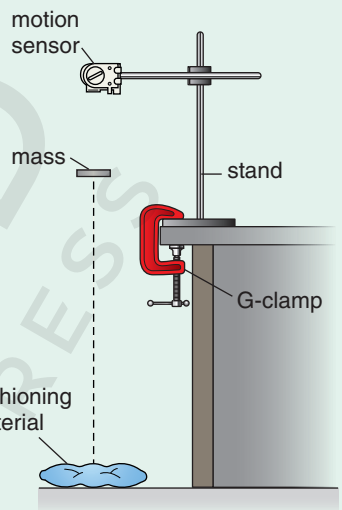
2 Start recording data. Release a mass at about 0.2 m below the sensor to allow it to fall freely onto the cushioning material.

'Release' usually means that the object falls from rest.

3 Stop recording data. The v–t graph obtained is a straight line, showing that the mass falls with a constant acceleration. Find this acceleration by calculating the slope of the graph.

4 Repeat steps 2 and 3 with different masses.

The mass should be released directly below the motion sensor.



- Keep a safe distance from the falling mass.
- Stay clear from the cushioning material to avoid being hit by the mass if it bounces off the cushion.

Discussion

1 What is the value of g obtained in the experiment?

2 State a source of error in the experiment.

Cross-link

The value of g at positions far away from the Earth will be discussed in Ch 10.

From Experiment 2c, you may find that the value of g is about 10 m s^{-2} (Fig 2.4c).

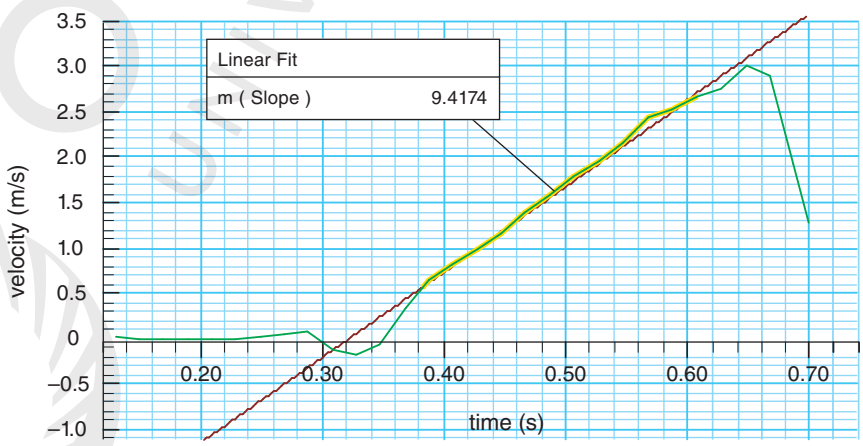


Fig 2.4c *A v–t graph obtained in Experiment 2c.*

In fact,

near the Earth's surface and in the absence of air resistance, all objects fall with the acceleration due to gravity g, which is 9.81 m s^{-2} downwards.

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Consider a ball falling freely from rest. It moves with a constant acceleration g and we can apply the equations of uniformly accelerated motion to study its motion. Figure 2.4d shows its motion and its motion graphs. The upward direction is taken as positive.

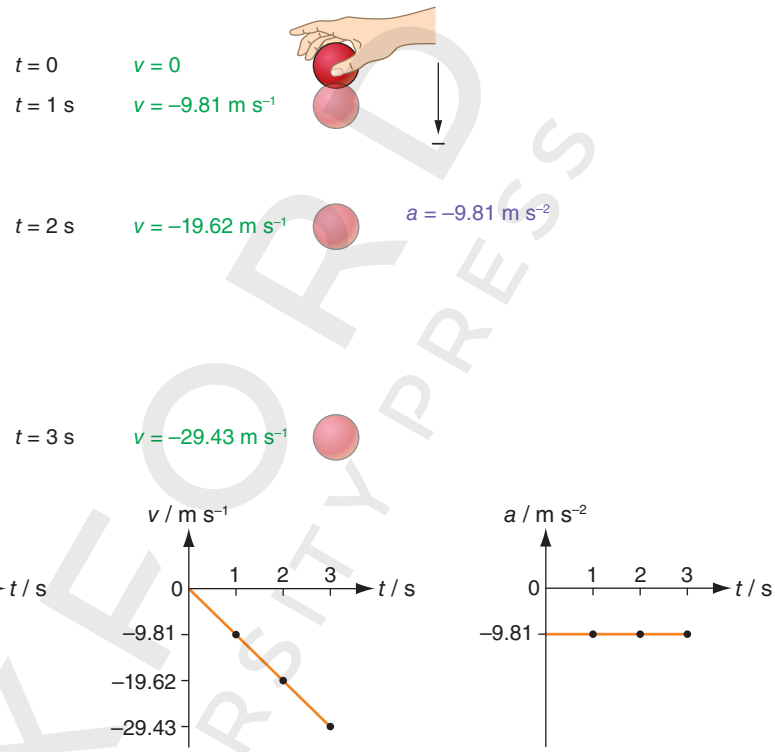
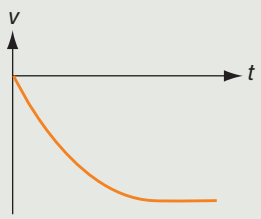


Fig 2.4d *A ball in a free fall.*

Know more

Falling in air

If the effect of air resistance is not negligible, an object will not fall with a constant acceleration. Its v–t graph will look like this:



This will be discussed in Ch 4.

STSE

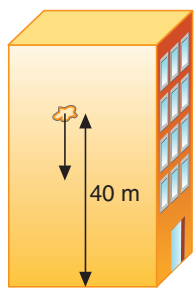
Falling objects can kill

The speed of the concrete in Example 11 reaches 28 m s^{-1} (≈ 101 km h^{-1}) when it falls from 40 m (higher than 10/F). Falling objects, even as light as a coin, can be very dangerous. They have caused many injuries in the past.



Example 11 Falling object

A piece of concrete falls from rest at a height 40 m above the ground from an old building. Neglect air resistance and take the acceleration due to gravity to be 9.81 m s^{-2}.



(a) How long does the concrete take to reach the ground?

(b) What is the speed of the concrete when it reaches the ground?

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Solution

The problem can be solved in two different ways.

Take upwards as positive.	Take downwards as positive.
(a) $u = 0$, $s = -40$ m, $a = -9.81$ m s^{-2}, t = ?	(a) $u = 0$, $s = +40$ m, $a = +9.81$ m s^{-2}, t = ?
By $s = ut + \frac{1}{2}at^2$,	By $s = ut + \frac{1}{2}at^2$,
$-40 = 0 + \frac{1}{2} \times (-9.81) \times t^2$	$40 = 0 + \frac{1}{2} \times 9.81 \times t^2$
$t = 2.86$ s or -2.86 s (rejected)	$t = 2.86$ s or -2.86 s (rejected)
The concrete takes 2.86 s to reach the ground.	The concrete takes 2.86 s to reach the ground.
(b) $u = 0$, $s = -40$ m, $a = -9.81$ m s^{-2}, v = ?	(b) $u = 0$, $s = +40$ m, $a = +9.81$ m s^{-2}, v = ?
By $v^2 = u^2 + 2as$,	By $v^2 = u^2 + 2as$,
$v^2 = 0^2 + 2(-9.81)(-40)$	$v^2 = 0^2 + 2 \times 9.81 \times 40$
$v = 28.0$ m s^{-1} (rejected) or -28.0 m s^{-1}	$v = 28.0$ m s^{-1} or -28.0 m s^{-1} (rejected)
The speed of the concrete is 28.0 m s^{-1} when it reaches the ground.	The speed of the concrete is 28.0 m s^{-1} when it reaches the ground.

Why is the answer $t = -2.86$ s rejected?

Use the values given in the question instead of those calculated from the previous part if possible to avoid error caused by rounding.

Why is the answer $v = 28.0$ m s^{-1} rejected when taking upward as positive and the answer $v = -28.0$ m s^{-1} rejected when taking downward as positive?

From Example 11, we can see that no matter which direction is taken as positive, the same numerical answer with the correct direction can be obtained by using the equations of uniformly accelerated motion.

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 A leaf falls more slowly than an apple because

A the leaf is not affected by gravity.

B the effect of gravity on the leaf is smaller.

C there is air resistance.

2 *True or false*: When a ball falls from a certain height, its acceleration increases as it falls faster and faster. (T/F)

Ss may wrongly think that a fast moving object must have a larger acceleration.

3 An object is released from rest from a height of 150 m. How long does it take to reach the ground? Complete the table below and solve the question twice by taking upward and downward as positive. Neglect air resistance. Take $g = 9.81$ m s^{-2}.

Take upward as positive	Take downward as positive
$s =$ _____ $a =$ _____ $u =$ _____	$s =$ _____ $a =$ _____ $u =$ _____
$t =$ _____	$t =$ _____

2 More about vertical motion under gravity

The acceleration due to gravity does not only apply on falling objects. Indeed,

> **when an object moves under the influence of gravity only, it always accelerates downward at 9.81 m s^{-2}.**

Consider a ball being thrown upward. The ball has an **initial upward velocity** and moves with a constant acceleration ***g* pointing downwards**. Figure 2.4e shows its motion and Figure 2.4f on p.77 shows its motion graphs. The upward direction is taken as positive.

Simulation

→ This simulation illustrates the motion of an object under gravity using *v*–*t* graphs. The object can be set to be released from rest, projected upwards and projected downwards under gravity. It is available on OUP web in PC version (.exe) only.

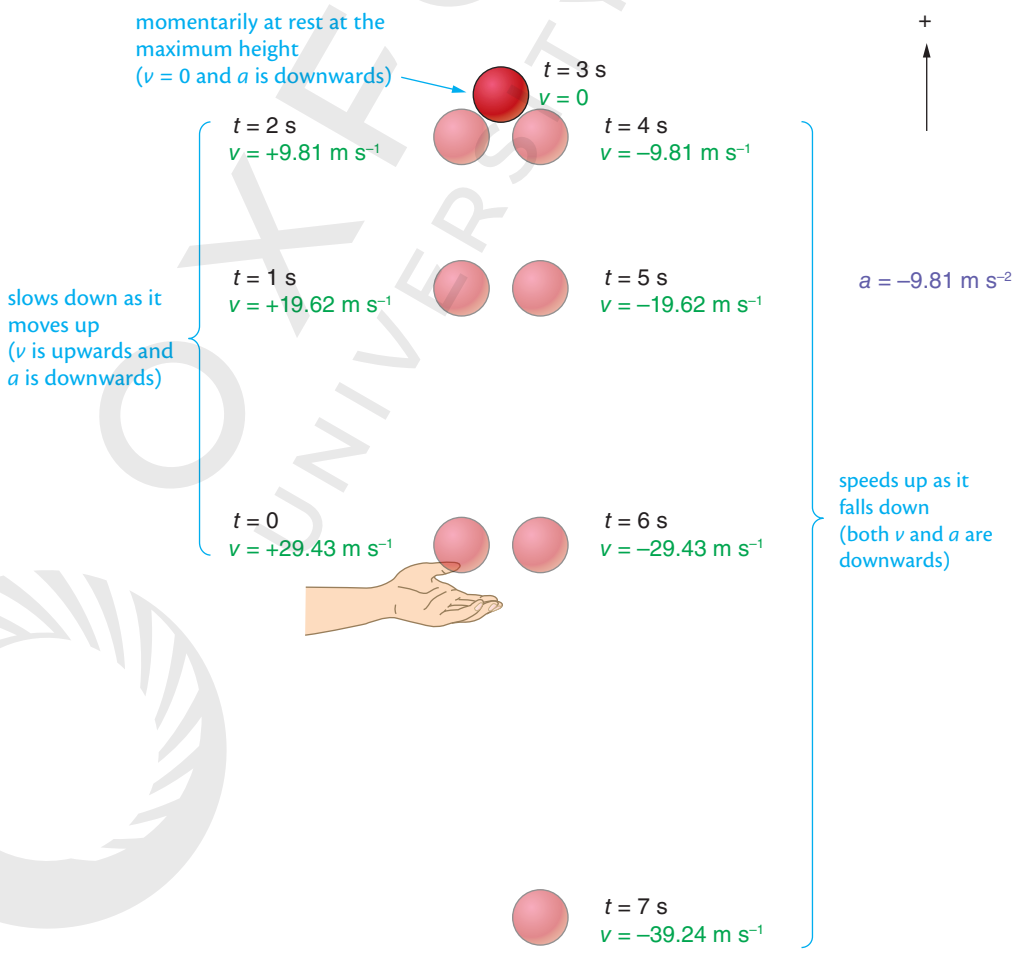


Fig 2.4e *Motion of a ball being thrown upward.*

Know more

Measuring *g* using apps

Open an app that can measure the acceleration of your mobile device. Hold your device to your body and jump as high as you can. You can find that the acceleration measured when you are moving in the air is about the same during upward or downward motion and the value is *g*.



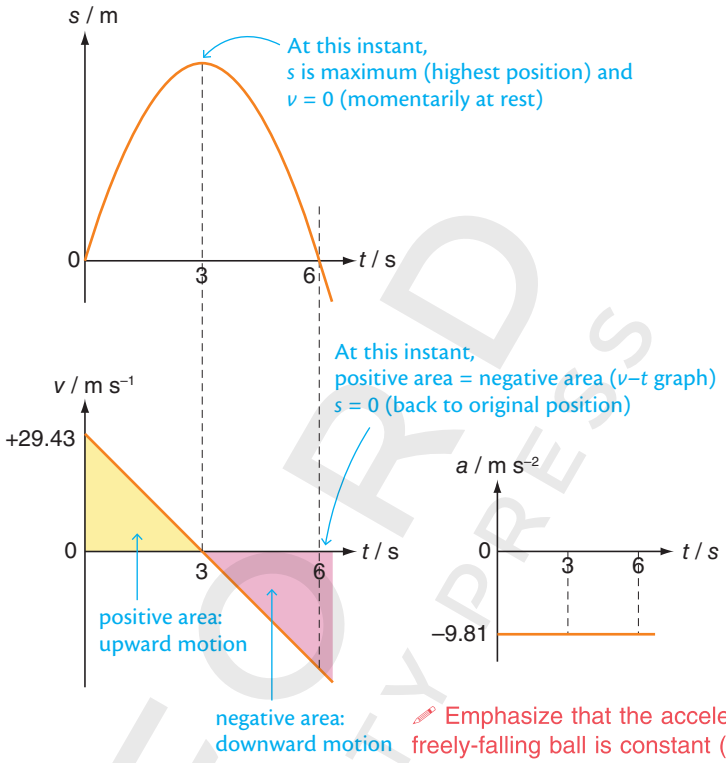


Fig 2.4f *Motion graphs of the ball.*

Emphasize that the acceleration of the freely-falling ball is constant (9.81 m s^{-2} downwards), although its velocity is changing both in magnitude and direction.

Note that the ball is **momentarily at rest at the highest position**. Also, the *v*–*t* graph of the ball is a straight line as the ball's acceleration is constant and points downwards when it is moving upward, moving downward or momentarily at rest.

Quick check 11

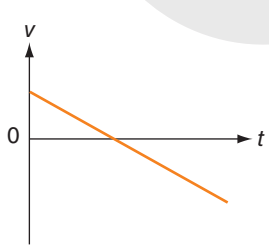
→ E-version in OUP Exercise Platform is also available.

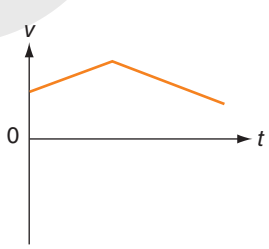
Google Forms Microsoft Forms

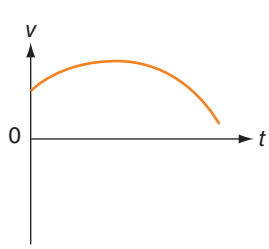
1 A ball is thrown upwards. Take upward as positive. Which of the following is/are correct? Put a tick in the corresponding box(es).

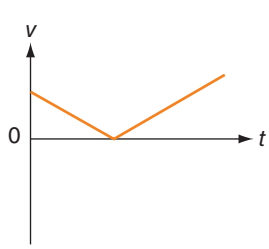
- ☐ During its upward motion, its velocity is positive.
- ☐ During its upward motion, its acceleration is positive.
- ☐ When it is at the maximum height, its velocity is zero.
- ☐ When it is at the maximum height, its acceleration is zero.
- ☐ During its downward motion, its velocity is negative.
- ☐ During its downward motion, its acceleration is negative.

2 Which of the following shows the *v*–*t* graph of an object being thrown upwards at $t = 0$? Take upward as positive. Neglect air resistance.

A 

B 

C 

D 

Example 12 Serving a tennis ball

At time $t = 0$, Simon throws a tennis ball vertically upwards at 7 m s^{-1} from a height of 1.70 m above the ground. When the ball falls back to a height of 3.30 m, Simon hits it. Neglect air resistance and take $g = 9.81$ m s^{-2}.

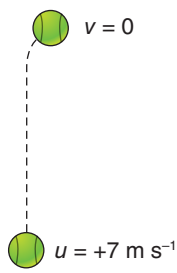


(a) What is the maximum height above the ground reached by the ball?

(b) When does Simon hit the ball?

Solution

Take upwards as positive.



(a) Consider the upward motion to the maximum height.

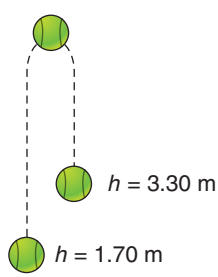
$u = +7$ m s^{-1}, $v = 0$, $a = -9.81$ m s^{-2}, $s = ?$

By $v^2 = u^2 + 2as$,

$0^2 = 7^2 + 2 \times (-9.81) \times s$

$s = 2.50$ m

Maximum height above the ground $= 1.70 + 2.50 = 4.20$ m



(b) Consider the motion from the starting position to the height 3.30 m via the maximum height.

$u = +7$ m s^{-1},

$a = -9.81$ m s^{-2},

$s = +(3.30 - 1.70) = +1.60$ m,

$t = ?$

By $s = ut + \frac{1}{2}at^2$,

$$1.60 = 7t + \frac{1}{2} \times (-9.81) \times t^2$$

$4.905t^2 - 7t + 1.60 = 0$

$$t = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4.905)(1.10)}}{2 \times 1.10}$$

$t = 1.14$ s or 0.286 s (rejected)

Simon hits the ball at $t = 1.14$ s.

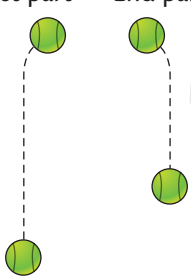
Skill *Mathematics*

The quadratic formula

▸ p.454

You may also solve this question by dividing the motion into two parts (upwards and downwards) and finding the time for completing each part of motion separately.

1st part 2nd part



Learn by practice 6

Take $g = 9.81$ m s^{-2} and neglect air resistance.

1 An object is thrown upwards with a speed x. Find the maximum height it can reach and the time it takes in each of the following cases.

(a) $x = 3$ m s^{-1}

(b) $x = 1.5$ m s^{-1}

2 An object is thrown upward with a speed x. Find the time it takes to return to the original level in each of the following cases.

(a) $x = 3$ m s^{-1}

(b) $x = 1.5$ m s^{-1}

3 A stone is thrown upwards from a cliff which is 50 m above the sea. If the initial speed of the stone is 5 m s^{-1}, after how long will it reach the sea?

4 A stone is released from rest from a cliff. If it reaches the sea in 5.5 s, how high is the cliff above the sea?

5 A stone is thrown downwards from a cliff which is 45 m above the sea. It reaches the sea after 2.5 s. Find the initial speed of the stone.

We have learned about the reaction time in chapter 1. Now you are able to estimate your own reaction time by using the equations of uniformly accelerated motion, as demonstrated by the following activity.

Video

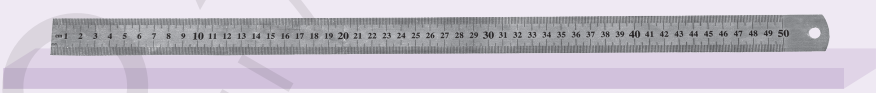
→ This video demonstrates the Hands-on activity.

Estimate your reaction time

Hands-on activity

What you need

50-cm ruler



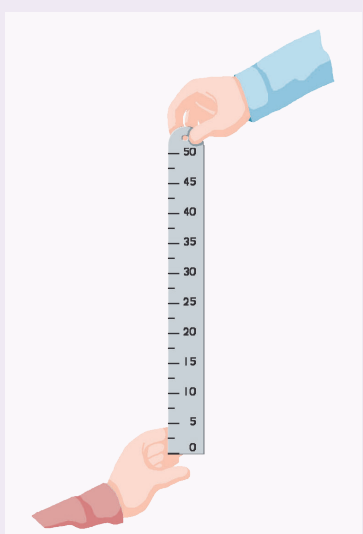
What to do

1 Ask your classmate to hold a 50-cm ruler vertically at the top end with the 0-cm mark at the bottom. Put your hand at the 0-cm mark without touching the ruler.

2 Ask your classmate to release the ruler at any time. You should catch the ruler when you see it starts falling. Record the mark where you catch the ruler.

3 Calculate your reaction time by applying

$$s = ut + \frac{1}{2}at^2.$$



What you find

What is your reaction time?

☆ HKDSE 2013 Paper 1A Q8

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Free fall

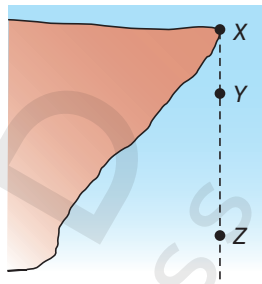
A stone falls from rest from point X on a cliff. The time it takes to fall from X to Y is T and from Y and Z is $3T$. Find $XY : YZ$.

A 1 : 3

B 1 : 9

C 1 : 15

D 1 : 16



Solution

Take downward as positive. Consider motion from X. $u = 0$, $a = +9.81\ \text{m s}^{-2}$

$s = ut + \frac{1}{2}at^2 = 4.905t^2 \propto t^2$

Common mistake

Students may not be able to use $s = \frac{1}{2}gt^2$ to solve this problem.

$\therefore \frac{s_1}{s_2} = \frac{t_1^2}{t_2^2}$

$\frac{XY}{XZ} = \frac{T^2}{(T + 3T)^2} = \frac{1}{16}$

Common mistake

Students may wrongly think that $\frac{XY}{YZ} = \frac{T^2}{(3T)^2}$. The initial speed u is not zero when the stone falls from Y to Z, so $s = \frac{1}{2}gt^2$ cannot be applied to find YZ.

If $XY = d$, then $XZ = 16d$.

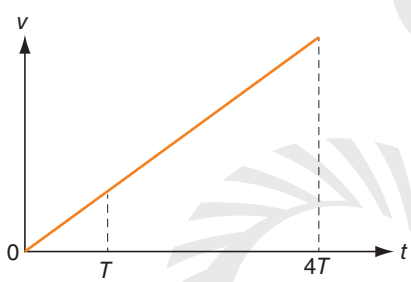
$YZ = 16d - d = 15d$

$XY : YZ = 1 : 15$

$\therefore$ The answer is C.

Alternative method:

The v–t graph of the stone is as shown below.



$\frac{XY}{XZ} = \frac{\text{area under graph from 0 to } T}{\text{area under graph from 0 to } 4T} = \frac{T^2}{(4T)^2} = \frac{1}{16}$

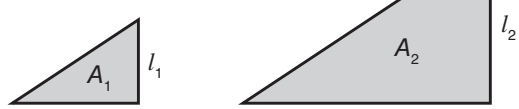
The regions under the graph from 0 to T and from 0 to $4T$ are similar triangles. For similar shapes, $\frac{A_1}{A_2} = \left(\frac{l_1}{l_2}\right)^2$

If $XY = d$, then $XZ = 16d$.

$YZ = 16d - d = 15d$

$XY : YZ = 1 : 15$

$\therefore$ The answer is C.



▶ Revision 2 Q12 (p.86)

Exercise 2.4

Take $g = 9.81$ m s^{-2} and neglect air resistance.

★**1** Ball X is of the same size as ball Y but is much heavier. They are released from rest at the same time from a certain height. Which of the following statements about the balls is/are correct?

(1) X reaches the ground before Y.

(2) The velocity of each ball increases uniformly.

(3) Their accelerations are zero when they are just released.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**2** A kangaroo jumps up and returns to the ground after 0.5 s. What is the maximum height that it reaches?

A 0.307 m B 0.5 m

C 1.23 m D 2.45 m

★**3** An object falls from rest at time $t = 0$. It falls a distance s from $t = 0$ to $t = T$. What is the distance that it falls from $t = 0$ to $t = 2T$?

A $2s$ B $3s$

C $4s$ D $9s$

★**4** A stone is projected upwards at a speed v from the ground. Its time of travel in the air is t. If it is projected at a speed $2v$, what will its time of travel be?

A t B $\sqrt{t}$

C $2t$ D $4t$

★**5** A ball is thrown vertically at 10 m s^{-1}. Should the ball be thrown upwards or downwards so that it will attain a higher speed when it reaches the ground?

A Upwards

B Downwards

C The ball will attain the same speed in either case.

D This cannot be determined as the height of the ball when it is thrown is unknown.

6 A girl jumps in a sack (Fig a). The highest position she reaches is 0.3 m above the ground.



Fig a

What is the girl's velocity when

(a) she leaves the ground,

(b) she drops back to the ground?

7 NASA's Zero-G Research Facility is located underground in Ohio, USA (Fig b). It has a vertical drop tower located in a steel vacuum chamber. In this chamber, an experiment vehicle released from rest experiences a free fall of 132 m before it is stopped by a decelerator cart.



Fig b

(a) How long does the vehicle experience free fall in the facility?

(b) What environment does the facility provide to allow the vehicle to experience free fall?

★**8** Johnny goes diving. He falls from rest at $t = 0$ and passes point A at 2 m s^{-1}. He reaches the water at $t = 0.8$ s.

(a) What is his initial height above the water?

(b) What is the height of A above the water?

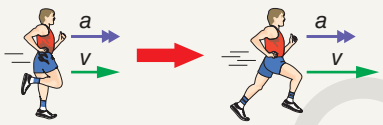
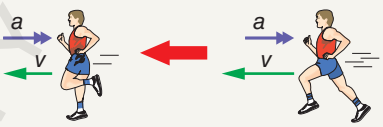
★**9** In the **Hands-on activity** on p.79, Ashely's hand is initially placed next to the 0-cm mark and she catches the rule at its 47-cm mark after the rule has been released. Estimate her reaction time.

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Key concepts 2

1 Describing motion along a straight line

- Direction of vectors (s, v, a…) along a straight line can be represented using positive (+) and negative (–) signs.
- The sign of acceleration of an object does not give the direction of motion, or whether the object is speeding up or slowing down.

speeding up (a and v have the same sign)		
slowing down (a and v have opposite signs)		

2 Motion graphs

- Displacement–time (s–t) graph
  - slope = velocity
  - sign of slope gives direction of velocity
- Velocity–time (v–t) graph
  - slope = acceleration
  - sign of slope gives direction of acceleration
  - area = displacement
  - sign of area gives direction of displacement
- Acceleration–time (a–t) graph
- Relation between motion graphs
  - s–t graph $\underset{\text{area}}{\overset{\text{slope}}{\rightleftarrows}}$ v–t graph $\xrightarrow{\text{slope}}$ a–t graph

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3 Equations of uniformly accelerated motion

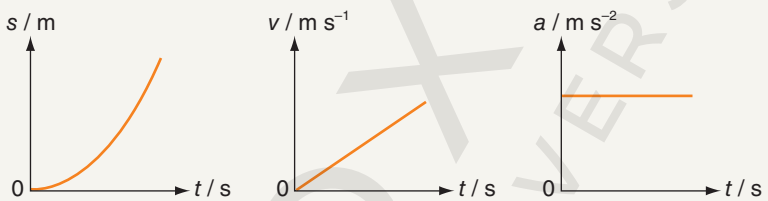
- $v = u + at$
- $s = \frac{1}{2}(u + v)t$
- $v^2 = u^2 + 2as$
- $s = ut + \frac{1}{2}at^2$

4 Free fall motion

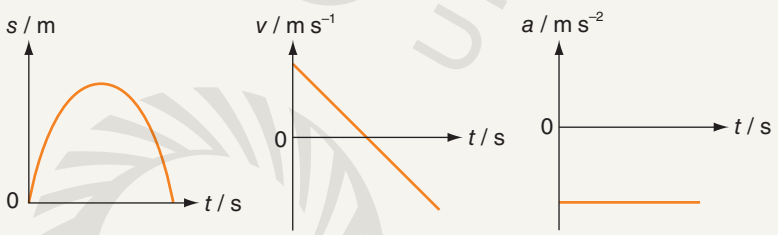
- Near the Earth's surface and in the absence of air resistance,
  - all objects fall with the same acceleration called acceleration due to gravity g
  - $g = 9.81\ \text{m s}^{-2}$ (downwards)



- Motion graphs of an object falling freely from rest (downwards as positive)



- Motion graphs of an object projected vertically upwards (upwards as positive)



Key terms

1 acceleration due to gravity 重力加速度 p.72
2 acceleration–time graph 加速度—時間關係線圖 p.55
3 air resistance 空氣阻力 p.72
4 displacement–time graph 位移—時間關係線圖 p.43
5 equations of uniformly accelerated motion 勻加速運動方程 p.63
6 free fall 自由下落 p.72
7 gravity 重力 p.72
8 velocity–time graph 速度—時間關係線圖 p.48

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Revision 2

Take $g = 9.81$ m s^{-2} and neglect air resistance.

Multiple-choice questions

★1 Misc A bus accelerates uniformly along a straight road. Figure a shows how its velocity v varies with its displacement s from a reference point.

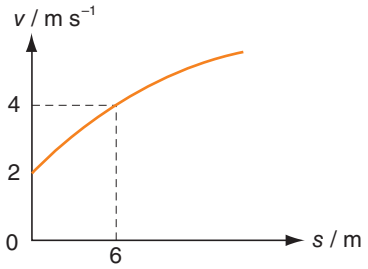


Fig a

What is the acceleration of the bus?

A 0.333 m s^{-2} B 0.666 m s^{-2}

C 1 m s^{-2} D 2 m s^{-2}

★2 2.3 A cat sees a mouse 2 m ahead. They start to accelerate uniformly to the right from rest at the same time. The cat catches the mouse after running 4 m (Fig b).

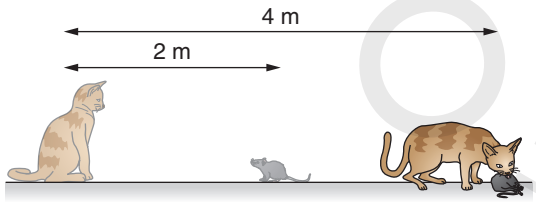


Fig b

If the acceleration of the mouse is 0.5 m s^{-2}, what is the acceleration of the cat?

A 0.5 m s^{-2} B 1 m s^{-2}

C 1.5 m s^{-2} D 2 m s^{-2}

★3 2.4 An object, initially at a height $2H$, falls from rest to the ground. The total time taken is $2T$. Its velocity is v_1 after falling a distance H. Its velocity is v_2 after falling for time T. Which of the following correctly shows the relation between v_1 and v_2?

A $v_1 > v_2$ B $v_1 = v_2$

C $v_1 < v_2$ D Cannot be determined

★4 2.2 Figure c shows the v–t graph of an object moving vertically.

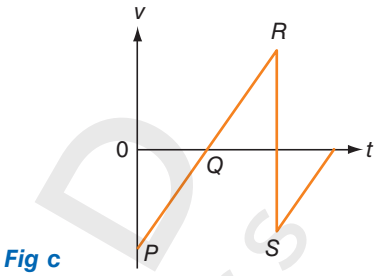


Fig c

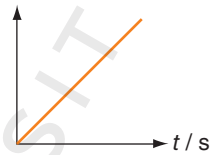
Which point refers to the moment that the object is furthest away from the starting position?

A P B Q

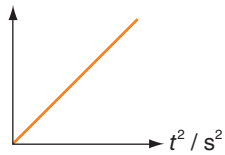
C R D S

★5 2.3 Which of the following graphs describe(s) an object accelerating uniformly from rest?

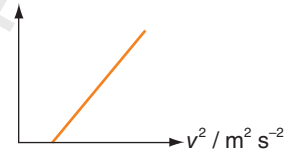
(1)



(2)



(3)



A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★6 Misc Figure d shows the v–t graph of a car moving along a straight road with a uniform acceleration.

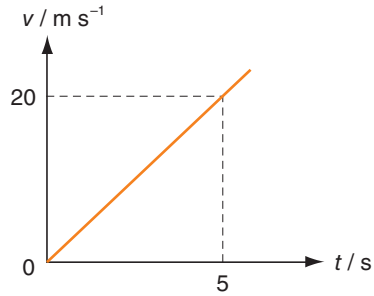


Fig d

How long does the car take to travel 30 m from the position at $t = 0$?

A 3 s B 3.87 s

C 4.23 s D 5 s

★7 2.3 A cheetah chases a prey along a straight path (Fig e).



Fig e

At time $t = 0$, they are 10 m apart and both of them accelerate uniformly from rest. The cheetah accelerates at 4 m s^{-2} and catches the prey at $t = 5$ s. Find the speed of the prey at $t = 5$ s.

A 4 m s^{-1} B 8.94 m s^{-1}

C 16 m s^{-1} D 20 m s^{-1}

★8 2.4 A diver jumps up from a springboard and then dives into the water (Fig f).

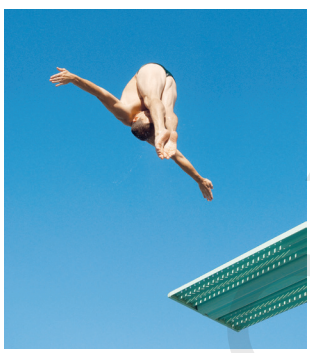
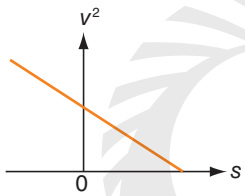


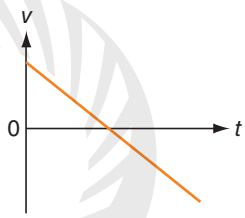
Fig f

Which of the following graphs correctly describe(s) the diver's motion?

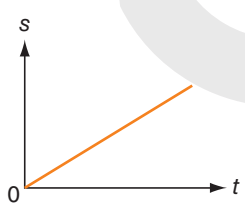
(1)



(2)



(3)



A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

★★9 Misc A car initially travelling at 30 m s^{-1} slows down uniformly along a straight road to stop in front of a traffic light. Its s–t graph is shown in Figure g.

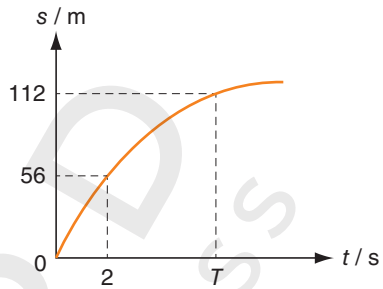


Fig g

What is T?

A 3.73 s B 4.00 s

C 4.37 s D 25.6 s

Refer p.64

★★10 2.3 In a traffic accident, a car leaves skid marks of 20 m long on the road during sudden braking. Investigators find that the same car travelling at 50 km h^{-1} leaves skid marks of 10 m long during sudden braking. Assume that the car brakes at the same constant deceleration in both cases. Estimate the speed of the car just before braking in the accident.

A 19.6 km h^{-1}

B 70.7 km h^{-1}

C 75 km h^{-1}

D 100 km h^{-1}



Refer p.64

★★11 2.3 An object accelerates uniformly from rest from point P. It moves along a straight line and passes points Q and R, with $PQ = QR$ (Fig h).



Fig h

If it takes time T to move from P to Q, how long does it take to move from Q to R?

A $(\sqrt{2} - 1)T$ B $0.5T$

C T D $\sqrt{2}T$

Refer p.80

12 **HKDSE 2013 Paper 1A Q8**
2.4

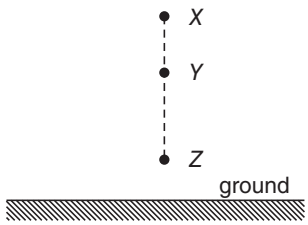


Fig i

A particle is released from rest at X as shown. It takes time t_1 to fall from X to Y and time t_2 to fall from Y to Z. If $XY : YZ = 9 : 16$, find $t_1 : t_2$. Neglect air resistance.

A 2 : 3 (10%)
B 3 : 4 (37%)
C 4 : 3 (12%)
D 3 : 2 (41%)

Q12 Exam Report: It seems that candidates were not competent in using the formula $s = \frac{1}{2}gt^2$ to tackle this problem.

13 **HKDSE 2014 Paper 1A Q5**
2.3

A particle is moving along a straight line with uniform acceleration. It takes 4 s to travel a distance of 36 m and then 2 s to travel the next 36 m. What is its acceleration?

A 2.5 m s^{-2}
B 3.0 m s^{-2} (37%)
C 4.0 m s^{-2}
D 4.5 m s^{-2}

14 **HKDSE 2015 Paper 1A Q9**
2.2

The figure shows the velocity–time (v–t) graph of two cars P and Q travelling along the same straight road. At $t = 0$, the cars are at the same position.

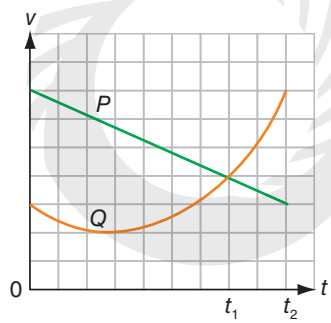


Fig j

Which deductions about the cars between $t = 0$ and $t = t_2$ are correct?

(1) P and Q are always travelling in the same direction.
(2) At $t = t_1$, the separation between P and Q is at a maximum.
(3) At $t = t_2$, Q lags behind P.

A (1) and (2) only (18%)
B (1) and (3) only (30%)
C (2) and (3) only (12%)
D (1), (2) and (3) (40%)

Q14 Exam Report: About one-third of the candidates did not realize that the separation between P and Q is at a maximum at $t = t_1$.

15 **HKDSE 2018 Paper 1A Q7**
Misc

Two blocks X and Y are moving towards the left. Their positions at successive instants (indicated by the numbers) of equal time intervals are shown below.

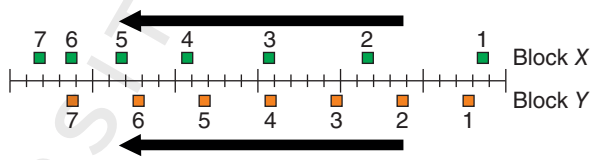


Fig k

Do these two blocks ever have the same instantaneous speed?

A Yes, at instant 3.
B Yes, at a moment between instants 4 and 5. (71%)
C Yes, at instant 6.
D No.

16 **HKDSE 2018 Paper 1A Q10**
2.3

A train departs from one station and stops at the next station after travelling 1.2 km along a straight line. The maximum acceleration and deceleration of the train are both 5 m s^{-2} and the highest speed of the train is limited to 20 m s^{-1}. Find the shortest time taken for this trip.

A 56 s
B 58 s
C 62 s
D 64 s (58%)

17 **HKDSE 2021 Paper 1A Q7**
2.2

At time $t = 0$, a small sphere is projected up along a smooth incline with a certain initial velocity. It travels a distance L and becomes momentarily at rest after a time $2T$. The corresponding velocity–time graph is shown below.

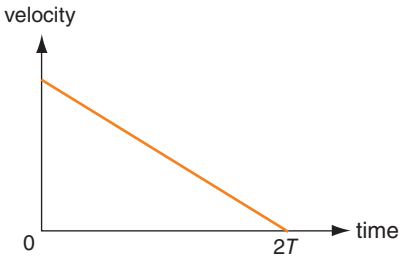


Fig l

What is the distance travelled by the sphere from $t = 0$ to $t = T$?

A $\frac{1}{4}L$

B $\frac{1}{2}L$

C $\frac{3}{4}L$ (58%)

D $\frac{4}{5}L$

Conventional questions

18 Misc
A ball is thrown upwards at a speed of 7 m s^{-1} from point X at $t = 0$. The ceiling is 2 m above X.

(a) When will the ball hit the ceiling? (2 marks)

(b) Sketch the v–t graph of the ball before it hits the ceiling. (2 marks)

***19** Misc
A speed boat with an initial velocity u speeds up uniformly at 4 m s^{-2} along a straight line. Its velocity is 25 m s^{-1} after travelling 30 m.

(a) Find u. (2 marks)

(b) Find the time taken for the above motion. (2 marks)

(c) Sketch the v–t graph for the above motion of the speed boat. (2 marks)

***20** 2.2
A car is travelling at 12 m s^{-1} along a straight road. At $t = 0$, the driver sees that a traffic light 17 m ahead turns from green to yellow and tries to stop the car. The v–t graph of the car is shown below (Fig m).

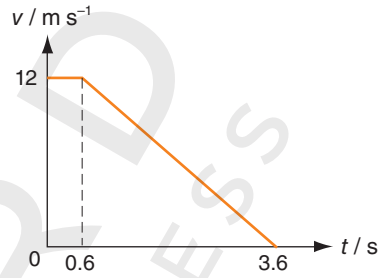


Fig m

(a) What is the reaction time of the driver? (1 mark)

(b) Find the acceleration of the car during braking. (2 marks)

(c) Can the car stop before the traffic light? Show your calculation. (3 marks)

***21** Misc
A trolley is given a sudden push and starts moving up an inclined runway from the lower end. A motion sensor is used to detect the motion of the trolley along the runway. Figure n shows the s–t graph obtained.

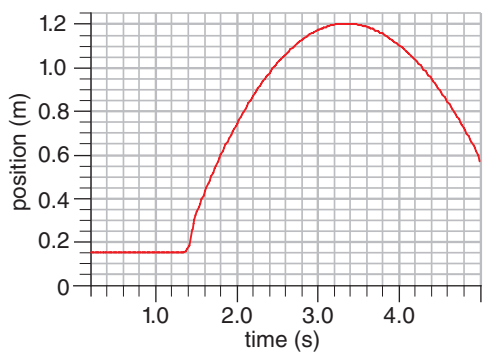


Fig n

(a) Is the motion sensor located at the upper end or lower end of the runway? (1 mark)

(b) Describe the motion of the trolley from $t = 2$ s to $t = 4.5$ s. (3 marks)

(c) It is known that the acceleration of the trolley is about constant from $t = 1.4$ s to $t = 4.5$ s. Estimate the magnitude of this acceleration. (2 marks)

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★22 Misc A ball is thrown upwards from point Y. The ball reaches the highest point X and then drops to Z before hitting the ground.

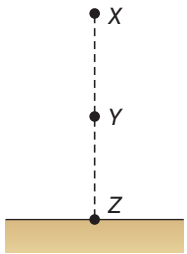


Fig o

(a) Sketch the s–t, v–t and a–t graphs of the ball. Take upwards as positive and Z as the reference point where $s = 0$. (6 marks)

(b) If X is taken as the reference point instead, which graph(s) in (a) will become different? (1 mark)

★23 Misc A monkey falls from rest for a height H (Fig p). The time taken for the fall is T.



Fig p

(a) (i) Sketch the v–t graph of the monkey during the fall. (2 marks)

(ii) What is the physical meaning of the area under the graph in (i)? (1 mark)

(b) (i) Express the final velocity V of the monkey in terms of H. (2 marks)

(ii) The monkey then falls a height of $2H$ from rest to another position. What is its final velocity after the fall? Express your answer in terms of V. (2 marks)

(c) A heavier monkey follows this monkey and falls a height $2H$ from rest. Compare its final velocity in the fall with the answer in (b)(ii). (1 mark)

★24 Misc Cars X and Y move along the same straight road. Car X moves at a constant velocity u. Car Y, which is initially 75 m behind car X, accelerates from rest with an acceleration of 2 m s^{-2}. It overtakes car X after 13 s.

(a) How far has car Y moved before it overtakes car X? (2 marks)

(b) Find u. (2 marks)

(c) Sketch the s–t graphs of cars X and Y in the same graph. (3 marks)

★25 Misc

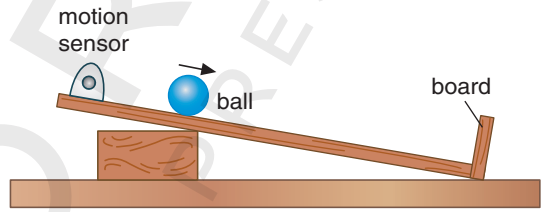


Fig q

A motion sensor is mounted at the higher end of an inclined runway. A ball held near the sensor is released from rest (Fig q). The ball collides with the board at the lower end of the runway and bounces many times before coming to rest. It keeps moving along a straight line for the whole process. Its v–t graph is shown below (Fig r).

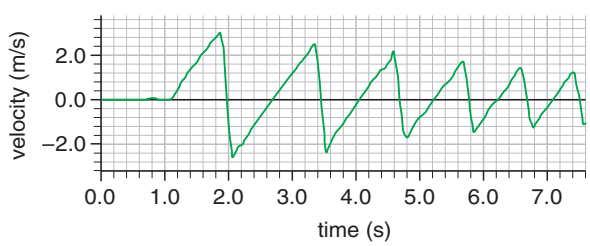


Fig r

(a) (i) When does the ball hit the board for the 1st time? (1 mark)

(ii) Estimate the speed of the ball at the moment in (a)(i). (1 mark)

(b) Estimate the acceleration of the ball at $t = 3.0$ s. (2 marks)

(c) Describe the motion of the ball from $t = 2.1$ s to $t = 3.3$ s. (3 marks)

(d) Estimate the distance between the initial position of the ball and the board. (2 marks)

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★**26** Misc An object is projected up an inclined plane at time $t = 0$. The object reaches the highest position and is momentarily at rest at $t = 2$ s. Then it moves down the plane and returns to the starting position at time T. Figure s shows its a–t graph.

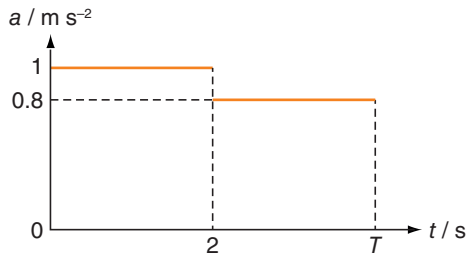


Fig s

(a) Find the object's initial velocity. (2 marks)

(b) Find T. (3 marks)

★**27** Misc A small ball, initially at rest, is released from a certain height to the ground and bounces several times. A motion sensor is used to detect its motion (Fig t). The s–t graph obtained is as shown (Fig u). Points A, B and C on the graph are at $t = 1.0$ s, 1.35 s and 1.7 s respectively.

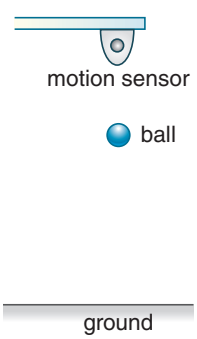


Fig t

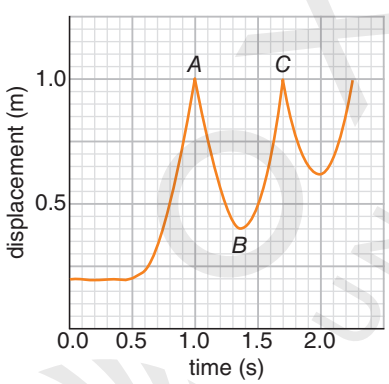


Fig u

(a) Which direction is taken as positive in the graph? (1 mark)

(b) Estimate the height of the ball above the ground when it is released. (1 mark)

(c) Describe the motion of the ball from $t = 1.0$ s to $t = 1.7$ s. (3 marks)

(d) Assume the air resistance is negligible.

(i) Sketch the v–t graph of the ball from $t = 1.0$ s to $t = 1.7$ s. (2 marks)

(ii) By considering the graph from $t = 1.35$ s to $t = 1.7$ s, estimate the acceleration due to gravity. (2 marks)

★**28** Misc Belinda jumps from a platform into the air. She takes 0.4 s to reach the maximum height and then falls into a swimming pool. The platform is 3 m above the water.

(a) What is Belinda's speed when she jumps from the platform? (2 marks)

(b) How long does Belinda stay in the air? (2 marks)

(c) Find Belinda's speed when she reaches the water. (2 marks)

(d) Sketch the v–t graph of Belinda during her motion in the air. (2 marks)

(e) Isabella jumps upwards at a speed doubled of that in (a).

(i) Sketch the v–t graph of Isabella in the same graph in (d). (2 marks)

(ii) If the maximum height above the platform reached by Belinda is h, what is the maximum height reached by Isabella? Express your answer in terms of h. (1 mark)

★**29** 2.4 Figure v shows an experiment set-up. When the switch is pressed, the ball is released from rest and a timer starts timing. When the ball hits the pad, the timer stops. A metre rule (not shown in Figure v) is available.

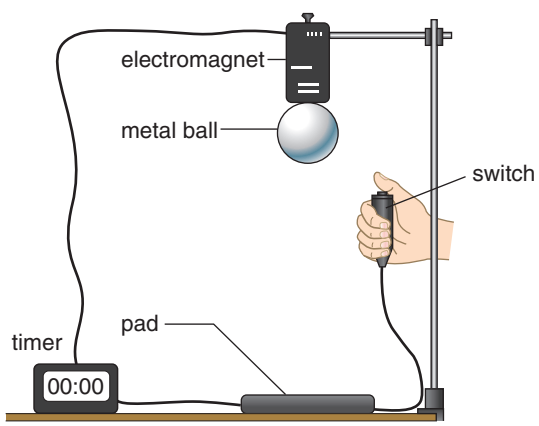


Fig v

Describe how you would use the apparatus to estimate the acceleration due to gravity. State clearly the physical quantities to be measured. (4 marks)

*30 Misc Read the following passage about **cat righting reflex** and answer the questions that follow.

Cats love to jump to high places and seem not afraid of falling down. They can turn quickly during a fall and land on their feet (Fig w). This ability is known as the righting reflex.

The minimum time needed for the turn in most cats is about 0.25 s. With this reflex, most cats can land safely from a height of several metres.

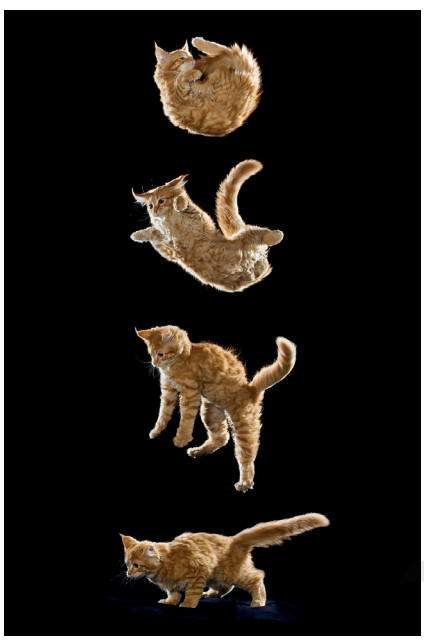


Fig w

(a) Estimate the minimum height needed for an average cat to right itself during a fall. (2 marks)

(b) Find the landing speed of a cat falling from rest from a height of 5 m. (2 marks)

(c) Sketch a graph to show how the landing speed v varies with the height of fall H. (2 marks)

**31 Misc A helicopter is rising upwards at a constant speed of 15 m s^{-1}. A sky diver leaves the helicopter at time $t = 0$ and the second diver leaves at $t = 1$ s. The velocity of each diver is the same as that of the helicopter at the moment of jumping.

(a) Find the distance between the first and the second diver at (i) $t = 1$ s, (ii) $t = 2$ s. (4 marks)

(b) The first diver opens his parachute when his speed reaches 66 m s^{-1}. When does he open his parachute? (2 marks)

(c) Sketch the v–t graphs of the first and the second divers from $t = 0$ to the moment in (b). (3 marks)

Refer p.78

**32 2.2 Cars X and Y are moving in the same straight lane on a highway. At time $t = 0$, car X brakes suddenly when it is 30 m in front of car Y. The driver of car Y applies the brake as soon as he is able to react. If the two cars do not collide with each other, their v–t graphs will be as shown in Figure x.

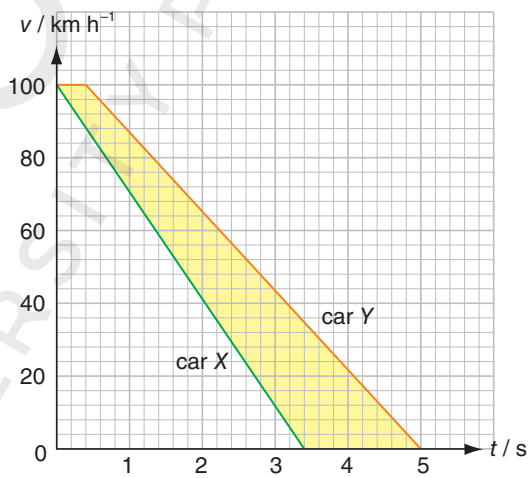


Fig x

(a) Find the decelerations of cars X and Y during braking. (3 marks)

(b) Find the distance which car Y has travelled during braking. (2 marks)

(c) (i) What is the physical meaning of the yellow area in Figure x? (1 mark)

(ii) By finding the yellow area, determine whether car Y will collide with car X. (2 marks)

(d) A driver should obey the two-second rule, which states that the distance between one's car and the car in front should not be shorter than the distance travelled by the car in two seconds. Does the driver of car Y obey this rule? (1 mark)

(e) If car X just slows down to 12 km h^{-1} instead of stopping, what would be the shortest distance between the two cars? (1 mark)

Refer p.54

33 HKDSE 2014 Paper 1B Q3
2.2

Two cars *A* and *B* initially at the same position, start to travel along the same straight horizontal road. The graph below shows how their velocities vary with time.

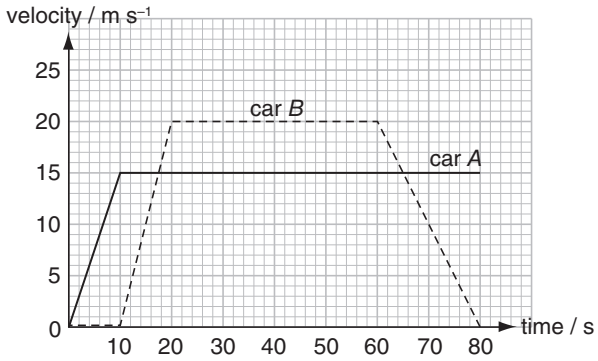


Fig y

(a) Describe the motion of car *A* along the whole journey from time $t = 0$ to $t = 80$ s. (2 marks)

(b) (i) Which car attained the greatest acceleration throughout the journey? Find this acceleration. (2 marks)

(ii) Sketch the acceleration–time (a–t) graph of car *B* from $t = 0$ to $t = 80$ s. (2 marks)

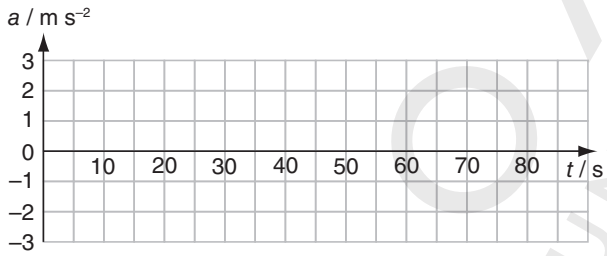


Fig z

(c) (i) At $t = 20$ s, what is the separation between cars *A* and *B*? (2 marks)

(ii) Deduce the time at which car *B* catches up with car *A*. (2 marks)

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2.4

A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure aa. Neglect air resistance.

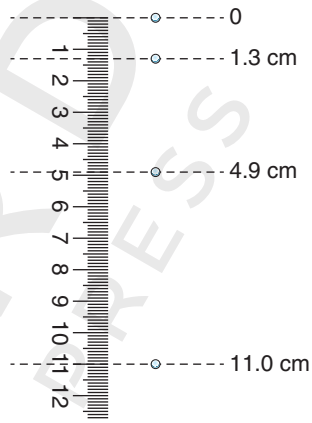


Fig aa

Estimate the acceleration due to gravity using the data in Figure aa. (2 marks)

Q33 Exam Report: This question was on the motion of two cars travelling along a straight horizontal road and was in general well answered.

(a) Most candidates described the motion of car *A* along the whole journey correctly.

(b) This part was well answered although some weaker candidates stated the unit of acceleration wrongly in (b)(i).

(c) (i) Most candidates understood that the area under a velocity–time graph represents displacement and were able to find the separation between cars *A* and *B*.

(ii) Only the more able ones deduced correctly the time at which car *B* caught up with car *A* using the result of (c)(i).

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Know more

Achilles and the tortoise

About 2400 years ago, Greek philosopher *Zeno* devised the following *paradox*.

Achilles races against a tortoise. His speed is 10 m s^{-1} and the tortoise's speed is 1 m s^{-1}. He allows the tortoise to start 100 m ahead of him. After he has run for 100 m, the tortoise is 10 m ahead. He then runs for 10 m and the tortoise is 1 m ahead. Whenever Achilles reaches the position the tortoise has been, the tortoise is still some distance ahead. He can never overtake the tortoise.



In this paradox, the time that Achilles needs to overtake the tortoise is divided into infinite time periods, each period being smaller than the previous one. It can be shown mathematically that the sum of these time periods is finite. That is, Achilles only needs finite time to overtake the tortoise.

Do you also think that Achilles cannot overtake the tortoise? You may draw their s–t graphs to find out the answer.

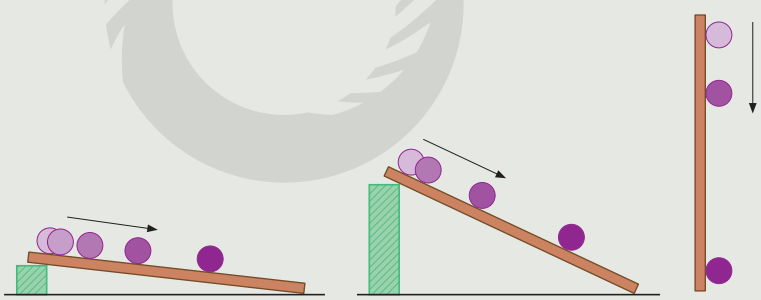
Know more

Galileo Galilei (1564–1642) (p.71)

Galileo Galilei was said to drop objects of various weights from the Tower of Pisa and compare their falls. Many people believe that he did not really do this. Instead, he did the following.

Since free fall motion was too fast to be measured accurately by any timing device available, Galileo used smooth inclined planes to 'slow down the motion'. He used a water clock for timing and found that balls of different weights rolled down the plane at the same rate. Also, the distance travelled by a ball was directly proportional to the square of the time.

He repeated the experiment with different inclined angles and got the same results. He argued that free fall was just motion along a plane inclined at 90° and should behave in the same way. This disproved the long-accepted idea that heavier object fell faster.



OUP eResources

Nature of science

Evolutionary nature of scientific knowledge

Galileo challenged the old theory on motion by using experiments and reasoning. This changed people's understanding of motion completely. This story shows that:

- Science is based on *empirical evidence*.
- Scientific knowledge is *evolutionary*. Even well-established scientific knowledge can be refuted or modified on the basis of new evidence. However, shifting people's belief from an old accepted concept to a new one may take time.
- Reasonable scepticism and the courage to challenge authority often leads to new scientific knowledge.

Zeno 芝諾　paradox 悖論　empirical evidence 經驗證據　evolutionary 演變的